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Photometric-weighted invariant feature transform for planetary surface image registration under complex illumination

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ABSTRACT

Accurate matching of planetary remote sensing images (PRSIs) is a critical prerequisite for planetary mapping, control network construction, three-dimensional reconstruction, and autonomous navigation and localization. However, under variations in imaging geometry and solar illumination, PRSIs often exhibit complex appearance changes, including brightness inversion, shadow migration, and nonlinear radiometric differences. These effects degrade local structural repeatability and descriptor consistency, leading to sparse reliable matches, increased outlier ratios, or registration failure. To address this problem, this paper proposes a photometric-aware invariant feature registration method, termed the Photometric-Weighted Invariant Feature Transform (PWIFT). PWIFT uses illumination-geometry-derived pixel-level photometric reliability as a soft prior throughout feature construction and matching. The method constructs a multi-scale photometric-weighted phase-congruency representation, guides keypoint selection toward reliable and repeatable terrain structures, and adopts a rotation-normalized bright-dark bi-channel descriptor with swap-aware matching to reduce ambiguity caused by brightness inversion and shadow-induced appearance changes. A coarse-to-fine rotation-scale search and homography-based reprojection cleanup are further incorporated to improve robustness to geometric perturbations and residual outliers. Experiments are conducted on a lunar multi-illumination image registration benchmark covering real illumination variations and controlled rotation-scale perturbations. Compared with representative conventional and learning-based methods, PWIFT achieves the highest SR@5 of 97.93%, with an ME@10 of 2.66 pixels and an RCM@5 of 90.80%. Additional analyses of illumination dependence, terrain dependence, matching density–accuracy trade-offs, and ablation results further demonstrate the stability of the proposed design. Moreover, PWIFT is validated on real Ryugu-ONC small-body observation images, indicating its applicability to robust PRSI registration under complex illumination and geometric variations.

1. Introduction

Surface image registration of planetary bodies is a fundamental component of deep-space exploration tasks, including planetary mapping, digital elevation model generation, three-dimensional reconstruction, cross-temporal change detection, autonomous navigation, and terrain-relative localization [1–5]. During close-range planetary mapping, spacecraft attitude, viewing geometry, and illumination conditions may vary considerably across observations. Meanwhile, planetary surfaces usually lack regular man-made structures and often exhibit weak textures, repetitive geomorphic patterns, and low local contrast. Therefore, reliable registration relies heavily on stable and repeatable terrain structures [6]. However, variations in solar incidence direction can cause shadow migration, penumbra displacement, brightness inversion, and terrain-dependent contrast changes, which substantially alter local intensity distributions and edge responses [7,8]. These effects make it difficult to establish stable homologous correspondences across multi-illumination planetary images.

Existing planetary image matching methods mainly include template-based, feature-based, structure-domain, and learning-based approaches [9–13]. Template-based methods are sensitive to illumination differences, occlusions, rotation, and scale changes. Classical sparse features rely on repeatable local extrema and gradient consistency, which may be weakened by strong shadows and nonlinear radiometric differences. Structure-domain and phase-domain methods reduce dependence on intensity magnitude, but pseudo responses may still appear near deep shadows, invalid regions, and penumbra boundaries. Learning-based matchers can exploit stronger contextual priors and

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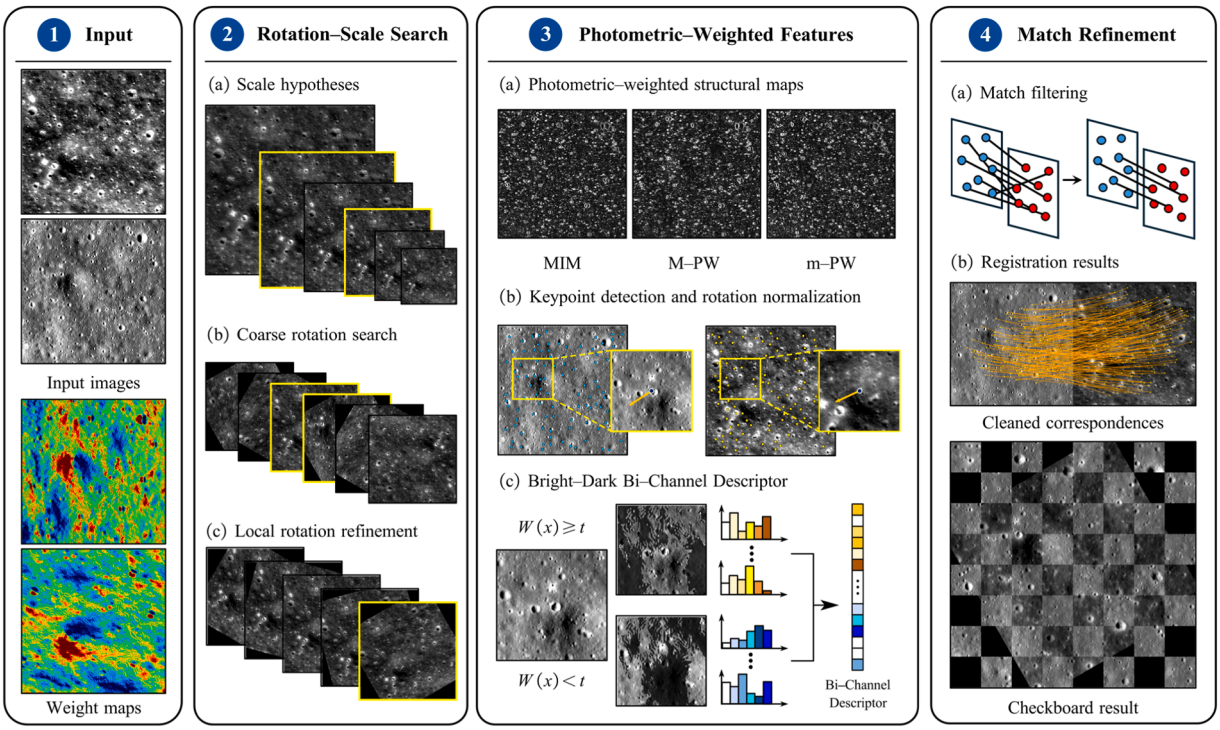

Fig. 1. Overall workflow of PWIFT.

generate abundant correspondences, but their generalization may be affected by the gap between common training domains and planetary images in terrain morphology, noise distribution, shadow geometry, and illumination mechanisms.

These limitations indicate that the key challenge in planetary multi-illumination image registration is not merely global brightness variation, but the spatially nonuniform reliability of local structures induced by illumination geometry. Some weakly illuminated or near-terminator regions may still contain useful geometric structures, whereas over-bright, deeply shadowed, or invalid regions may produce unstable responses. In addition, brightness inversion and shadow migration can change local orientation statistics and descriptor distributions, and these appearance changes may be further coupled with rotation-scale variations. In this study, we hypothesize that photometric reliability derived from illumination geometry can serve as an interpretable prior for the local repeatability of planetary terrain structures. Based on this hypothesis, we propose PWIFT, a photometric-aware registration method that embeds pixel-level reliability into multi-scale phase-congruency representation, keypoint selection, descriptor construction, and homography-based reprojection cleanup. Specifically, PWIFT constructs a photometric reliability map from illumination geometry, performs photometric-weighted structural feature extraction, introduces a rotation-normalized bright-dark bi-channel descriptor with swap-aware matching, and incorporates a coarse-to-fine rotation-scale search with homography-based cleanup. Fig. 1 shows the overall workflow of PWIFT. The main contributions of this paper are summarized as follows.

1) A photometric-reliability-guided planetary image registration framework, termed PWIFT, is proposed for complex multi-illumination scenarios. It converts illumination geometry into a pixel-level soft reliability prior and embeds it into the matching process, rather than using photometric information only for image correction or preprocessing.
2) A multi-scale photometric-weighted structural feature construction strategy is developed. By combining phase-congruency-based structural representation, soft reliability weighting, and photometric-weighted keypoint selection, PWIFT reduces the influence of unreliable shadowed, over-bright, and invalid regions while retaining useful terrain structures.
3) A bright-dark bi-channel descriptor and swap-aware matching strategy are introduced to improve descriptor consistency under brightness inversion and shadow migration. A coarse-to-fine rotation-scale search and homography-based reprojection cleanup are further incorporated to enhance robustness under rotation, scale, and coupled geometric perturbations.
4) A comprehensive evaluation is conducted on a lunar multi-illumination benchmark and an additional small-body registration test. The experiments cover illumination variations, controlled geometric perturbations, density–accuracy trade-offs, ablation analysis, and comparisons with representative conventional and learning-based methods.

The rest of this paper is organized as follows. Section II reviews related work. Section III presents the proposed PWIFT framework. Section IV reports the experimental results and analyses. Section V concludes this paper.

2. Related work

PRSIs often exhibit weak textures, repetitive patterns, strong illumination-geometry variations, and nonlinear radiometric differences. They are also subject to engineering constraints, including ultra-large strip images, irregular overlap regions, and cross-track or cross-sensor observations [14]. These factors make it difficult to establish stable homologous correspondences under severe appearance changes. Existing methods can be broadly categorized into template-based matching, feature-based matching, and learning-based matching.

2.1. *Template-based matching*

Template-based matching typically slides a template or window within a predefined search range and determines correspondences using a similarity metric. It is straightforward to implement and can achieve high accuracy when local geometric deformation is limited. To handle

scale, rotation, and imaging-geometry variations in planetary images, some studies introduce network constraints or global consistency to improve stability. For example, a triangulation-network-constrained strategy for automatic full-moon image registration can enhance global geometric consistency and reduce the accumulation of local mismatches to some extent [15]. Related terrain-aided studies have also combined template matching, terrain reconstruction, DEM enhancement, or terrain-aware super-resolution to improve local matching reliability under uncertain observations [16,17]. However, area-based methods remain sensitive to illumination differences, occlusions, strong shadow boundaries, and rotation-scale changes. When valid regions are fragmented or radiometric discrepancies are significant, it remains difficult to construct a reliable similarity measure.

2.2. Feature-based matching

Classical sparse features perform well in terrestrial scenarios, but their implicit assumption of consistent intensity or gradients can degrade under strong shadows and pronounced radiometric differences. To improve robustness across illumination and radiometric conditions, planetary and multimodal remote sensing image matching studies have adopted structural-domain or phase-domain representations to reduce dependence on intensity magnitude and local gradient consistency [18]. Recent benchmark studies further show that feature-based methods remain important for multimodal remote sensing image matching under nonlinear radiometric differences and cross-modal appearance variations [19]. Representative examples include efficient feature transforms built on phase congruency for fast PRSI matching [11], bandpass filtering with local topological priors [20], global-to-local invariant feature transformation [21], sparse-sampling-based structural description [22], and modality-invariant structural representation for Earth and planetary remote images [23]. For multi-illumination orbital images of typical planetary bodies such as the Moon, crater-neighborhood structures have also been exploited to improve feature stability and outlier resistance under varying illumination [24].

Engineering-oriented improvements for high-resolution orbital imagery have also received attention, including image partitioning, fast local correspondence establishment, dual-channel aggregation feature transforms, and benchmark-based method evaluation [10,12,19,25]. Outlier rejection remains another critical step because repetitive terrain patterns and locally similar structures may generate geometrically inconsistent matches. Mismatch removal strategies that combine global constraints with local geometric preservation can improve the credibility of final correspondences [26]. Although these methods improve robustness through structural representation, descriptor invariance, or geometric filtering, most of them treat illumination variation implicitly through radiometrically robust features or post-processing constraints. The illumination geometry itself is rarely converted into a pixel-level reliability prior and propagated through feature response computation, keypoint selection, descriptor statistics, and geometric refinement.

2.3. Learning-based matching

In recent years, learning-based matchers have shown advantages in low-texture scenes, large baselines, and complex geometric changes. General image matching frameworks, such as SuperPoint [27] combined with LightGlue [28], LoFTR [29], XoFTR [30], DKM [31], and RoMa/RoMaV2 [32,33], exploit learned descriptors, attention-based context aggregation, detector-free matching, or dense correspondence estimation to improve matching robustness. Recent remote sensing studies have also introduced stronger semantic and structural priors, including SAM-guided focused attention for multimodal remote sensing image matching [34] and keypoint-free feature tracking for multisource SAR image motion retrieval [35]. Deep neural networks have also been used in aerospace-related structure identification tasks, such as weak stability boundary identification [36], reflecting the increasing use of data-driven models in remote sensing and space applications.

Learning-based methods have also been explored in planetary and deep-space scenarios, including lunar rover image matching and reconstruction [13], semantic-aided Mars rover reconstruction [37], mesh-guided small-body multi-view matching [38], and on-ground visual-navigation validation for orbital GNC [3]. Surveys of deep-space image-based navigation further indicate that robust matching and repeatable feature extraction under complex illumination remain key capabilities for mapping and navigation applications [6]. However, planetary imagery often exhibits substantial gaps from common training domains in surface textures, noise distributions, imaging geometry, and illumination conditions, which may lead to uncertain generalization and increased outlier ratios. Therefore, learning-based methods are important baselines for evaluation, but physically interpretable and training-free approaches remain valuable for planetary scenarios where labeled training data are limited. PWIFT is designed to address this gap by embedding illumination-geometry-derived photometric reliability into feature construction, keypoint selection, descriptor matching, and homography-based reprojection cleanup.

3. Methodology

3.1. Overview of the PWIFT framework

Given a reference image I_r and a moving image I_m, PWIFT aims to obtain reliable correspondences and estimate the geometric relationship between the two images under complex illumination, brightness inversion, shadow migration, rotation, and scale variations. In addition to the image pair, the corresponding photometric weight maps and illumination masks are used to provide pixel-level reliability information. The final output is a cleaned correspondence set M_{clean} and an estimated homography $\widehat{H}_{r\to m}$.

The PWIFT pipeline consists of two coupled parts: photometric-reliability-guided feature construction and geometry-guided match refinement. First, photometric reliability maps are constructed from illumination geometry and are used to guide the multi-scale phase-congruency representation. At each pyramid level, PWIFT computes photometric-weighted structural maps, including M_{PW}, m_{PW}, and MIM. Keypoints are then selected from the structural maps using both structural saliency and photometric reliability, and a dominant orientation is estimated for local rotation normalization. Around each selected keypoint, a rotation-normalized bright-dark bi-channel descriptor is constructed to improve robustness to brightness inversion and shadow-induced appearance changes.

To handle large global rotation-scale variations, PWIFT performs a coarse-to-fine rotation-scale hypothesis search before the final full matching stage. For each candidate hypothesis, the moving image is transformed into a candidate frame, and a lightweight PWIFT matching process is performed to estimate its geometric consistency. The best rotation-scale hypothesis is selected according to lexicographic geometric-consistency ranking. Under the selected hypothesis, full PWIFT feature extraction, descriptor construction, and descriptor matching are performed. The moving-side matched points are then mapped back to the original coordinate system of I_m.

Finally, descriptor matches are obtained using a swap-aware distance between the bright and dark channels, followed by mutual check and Fast Sample Consensus (FSC) [39] geometric filtering. A homography-based reprojection cleanup is further applied to remove residual outliers and generate M_{clean}. The detailed formulations of each module are presented in the following subsections.

3.2. Photometric reliability modeling

In close-range planetary remote sensing, the same terrain element may undergo brightness inversion, shadow migration, and contrast drift

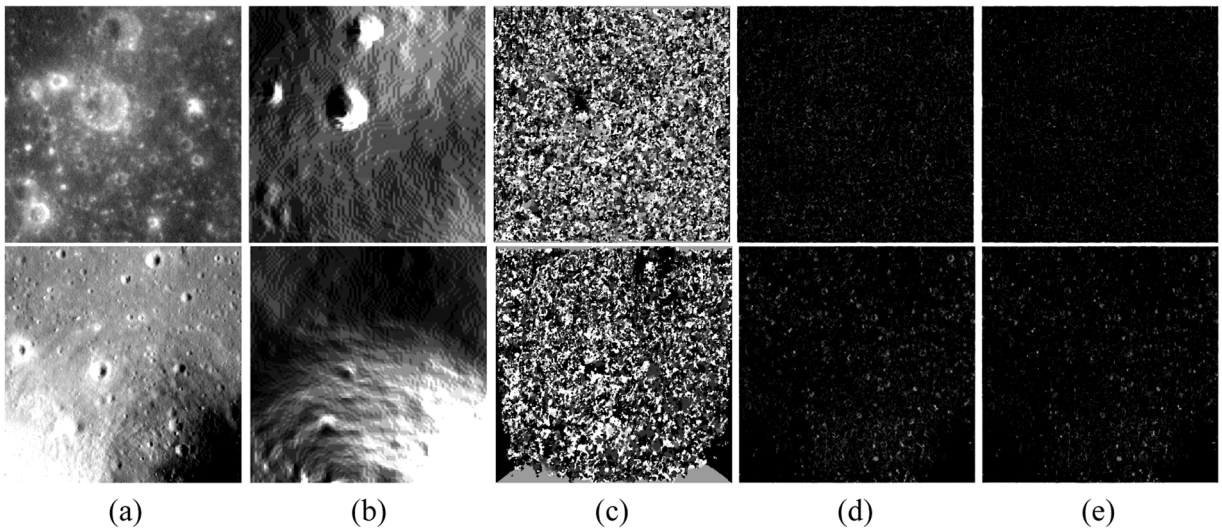

Fig. 2. Visualization of the photometric weights and feature map results. (a) Original image, (b) photometric weight map, (c) MIM, (d) M-PW, and (e) m-PW.

as imaging geometry and solar illumination change, which reduces the repeatability of local structures based only on intensity or gradients under extreme illumination. To explicitly separate photometrically reliable regions from geometrically visible but weakly illuminated regions in subsequent detection and description, a pixel-level photometric reliability map and an illumination mask are constructed for each image. The photometric reliability map is derived from interpretable illumination geometry. In this study, the Akimov disk function is adopted to model the attenuation of disk brightness with disk position and viewing configuration, so that deep shadows, near-terminator areas, and highly oblique viewing regions naturally receive lower reliability values.

Planetary image products usually provide either pixel-wise illumination geometry or radiometrically corrected images with auxiliary observation geometry. For the former case, the incidence angle i, emission angle e, phase angle α, and illumination mask S are directly read from the corresponding products. For the latter case, these quantities can be recovered from the observation epoch, camera model, and target-body shape model. Pixels with missing geometry, invalid surface intersection, background response, or non-illuminated status are excluded from the valid region and assigned zero reliability. This treatment avoids using unreliable observations in subsequent feature construction while retaining geometrically valid weak-illumination regions for soft weighting rather than hard rejection. Fig. 2 visualizes the photometric weight map and the corresponding structural feature maps. Compared with the original image, the photometric reliability map suppresses low-reliability regions caused by shadows, over-bright responses, and invalid observations, while the resulting MIM, M-PW, and m-PW maps show how photometric reliability affects the subsequent structural representation.

After the pixel-wise illumination geometry is recovered, the Akimov disk function is adopted as the core of photometric modeling to construct the photometric reliability map. Let the Akimov disk function be denoted as D_{Aki}. It is formulated as a function of the phase angle $\alpha$and the photometric coordinates (β,γ), which are determined by i, e, and the relative azimuth geometry $D_{\mathrm{Aki}}(\alpha,\beta,\gamma)$. The raw photometric weight for valid pixels is defined as

$$w_0(p) = D_{\mathrm{Aki}}\left(\alpha(p),\beta(p),\gamma(p)\right), p \in \Omega \tag{1}$$

Invalid pixels are then assigned $w_0(p) = 0$. To improve cross-image comparability and to avoid numerical scale differences caused by different observation configurations from affecting subsequent thresholding and weighting, w_0 is linearly normalized within the valid region Ω to obtain the final weight $w(p) \in [0,1]$:

$$w(p) = \frac{w_0(p) - \min_{q\in\Omega} w_0(q)}{\max_{q\in\Omega} w_0(q) - \min_{q\in\Omega} w_0(q) + \varepsilon} \tag{2}$$

where ε is a small constant to prevent a zero denominator. With this definition, pixels with moderate incidence and emission angles and stable illumination conditions receive higher reliability values, whereas pixels near deep shadows, the terminator, or geometrically unstable regions are downweighted.

It should be noted that the photometric reliability map W is not used as a hard mask, a simple radiometric correction result, or a post-hoc matching weight. Instead, it converts illumination geometry into a pixel-level soft reliability prior that is consistently propagated throughout the proposed PWIFT pipeline. Specifically, W is incorporated into the subsequent multi-scale phase-congruency representation, photometric-weighted keypoint scoring, dominant-orientation estimation, and bright-dark descriptor voting. In this way, photometric information is embedded into the feature construction process rather than being applied only as an external weighting factor. This design suppresses unreliable responses induced by extreme illumination, while preserving potentially useful structures in weakly illuminated regions, thereby providing a physically interpretable basis for robust matching under complex illumination.

3.3. Multi-Scale photometric-weighted phase-congruency representation

Close-range observations of planetary bodies often produce images with pronounced illumination nonuniformity. Under different imaging geometries, the same terrain unit can exhibit strong brightness inversion and contrast variations. These effects are especially evident near the transition between penumbra and deep shadow, where local intensity and gradient distributions are dominated by irradiance. As a result, conventional structural features based on intensity or gradients may become unstable. Phase congruency (PC) [18] emphasizes phase alignment and reduces the dependence on magnitude, and thus provides improved robustness to illumination changes. However, PC may still be affected by background responses, invalid pixels, and low-reliability shadowed regions. Moreover, when illumination variation is coupled with scale changes, a single-scale PC representation may be insufficient to maintain stable structural responses. Therefore, PWIFT constructs a multi-scale photometric-weighted PC representation to provide a stable basis for subsequent keypoint detection and descriptor construction.

Given an input image $I(x)$, a photometric weight map $W(x)$, and an illumination mask $L(x)$, synchronized pyramids are constructed for the

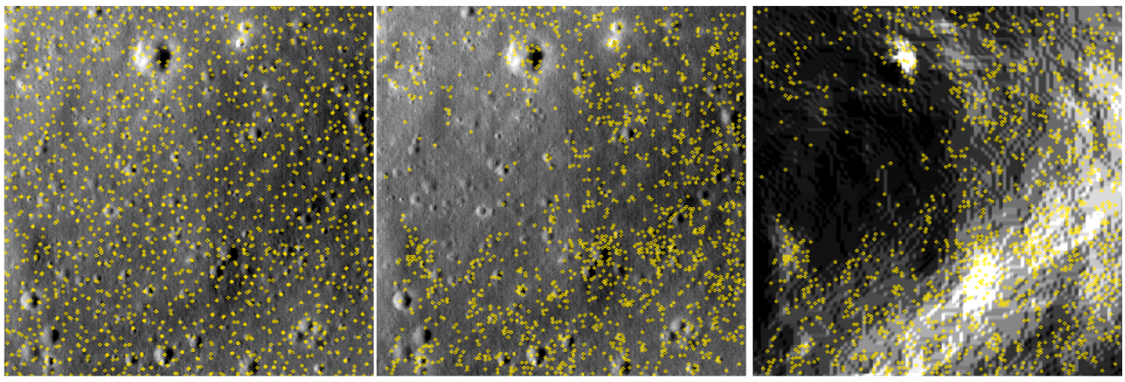

Fig. 3. Comparison of keypoint distributions before and after photometric weighting. (a) Keypoint distribution on the image without photometric weighting, (b) keypoint distribution on the image with photometric weighting, and (c) distribution of the photometrically weighted keypoints on the photometric weight map.

image, photometric reliability map, and illumination mask. At each pyramid level, I, W, and $L$are resized consistently, and the following PC computation is performed independently.

To reduce the influence of deep-space background and invalid pixels on frequency-domain filtering and directional energy statistics, a valid-pixel mask is constructed using $W$and L. Binary decisions are made using a background threshold w_{bg} and an illumination threshold on$_{thr}$, and take their union. This design avoids mistakenly discarding valid regions on the shadowed side when only W is used. The mask is defined as

$$m_W(x) = \left[W(x) \geq w_{bg}\right] \tag{3}$$

$$m_L(x) = \left[L(x) > on_{thr}\right] \tag{4}$$

$$mask(x) = m_W(x) \vee m_L(x) \tag{5}$$

where $[\cdot]$ denotes the binary indicator function. In implementation, simple morphological refinement can be applied to remove isolated noise and reduce boundary artifacts, while the main role of $mask(x)$ is to constrain PC computation to reliable or geometrically valid regions.

Within the valid region, a multi-scale and multi-orientation log-Gabor filter bank is applied to perform frequency-domain convolution, producing the even and odd responses at each scale and orientation. Following the model of Kovesi [18], the orientation-specific phase congruency $PC_o(x)$ and the accumulated orientation magnitude $CS_o(x)$ are computed. Noise-threshold suppression and frequency-spread weighting are retained so that $PC_o(x)$ concentrates more effectively on stable structural information. The orientation space is quantized into 12 bins, which provides a finer MIM representation for the subsequent dominant-orientation estimation and descriptor construction.

To reflect illumination reliability in the phase congruency results, a direct pixel-wise weighting of PC_o by W would uniformly reduce the phase congruency values in low-reliability regions. This reduction may weaken the structural contrast on the shadowed side and consequently degrade the discriminability of subsequent moment-based features. Instead, W is mapped into a piecewise linear soft weight $w_{soft}(x)$, which provides a smooth transition in low-weight regions and applies hard zeroing only to the background. The soft weight is defined as

$$w_{soft}(x) = clip\left(\frac{W(x) - w_{soft}^{lo}}{w_{soft}^{hi} - w_{soft}^{lo}}, 0, 1\right) \tag{6}$$

Based on this soft weight, the photometrically weighted orientation phase-congruency map is defined as

$$PC_o^{PW}(x) = \begin{cases} PC_o(x) \cdot w_{soft}(x)^{\gamma_{pc}} & , \quad m_W(x) \neq 0 \\ 0 & , \quad m_W(x) = 0 \end{cases} \tag{7}$$

where γ_{pc} controls the strength of the weighting. This soft-weighting strategy suppresses unreliable responses induced by extreme illumination, while preserving potentially useful structural information in weakly illuminated but geometrically valid regions.

Built on PC_o^{PW}, the standard phase-congruency covariance moment accumulation is performed to obtain orientation-independent moment features. The maximum moment $M_{PW}(x)$ and the minimum moment $m_{PW}(x)$ characterize the dominant and subordinate local structural energies, respectively. These quantities provide stable structural strength measures for subsequent keypoint detection and description. In addition, a maximum index map MIM is constructed by assigning each pixel the orientation index with the maximum accumulated orientation magnitude:

$$MIM(x) = \arg\max_o CS_o(x) \tag{8}$$

Finally, PC_o^{PW}, M_{PW}, m_{PW}, and MIM are computed at all pyramid levels. The resulting scale-specific structural maps are used for multi-scale keypoint detection, dominant-orientation estimation, and descriptor construction. Feature coordinates extracted from different pyramid levels are mapped back to the original image coordinate system before matching. In this way, PWIFT propagates the illumination-geometry-derived reliability prior across multiple structural scales, providing a more stable basis for matching under coupled illumination and scale variations.

3.4. Photometric-weighted keypoint detection and local rotation normalization

This section uses the moment features of M_{PW} and m_{PW} as structural baselines and incorporates the photometric reliability characterized by W as a soft constraint. Structural saliency and illumination availability are coupled into a unified keypoint selection framework, aiming to obtain a keypoint set that is repeatable, spatially well distributed, and robust to strong illumination differences.

Keypoint detection is performed on the photometric-weighted structural maps obtained at different pyramid levels. It consists of four stages: candidate extraction, photometric-weighted scoring, adaptive quality screening, and global spatial selection. Candidate extraction is performed on both M_{PW} and m_{PW} channels to cover dominant and subordinate structural information. The M_{PW} channel emphasizes dominant structural energy and mainly captures salient terrain boundaries and local corners, whereas the m_{PW} channel provides complementary candidates in weak-texture regions, brightness-inversion areas, or locations where the dominant structure is not sufficiently prominent. This dual-channel extraction improves structural diversity and reduces the risk of relying only on high-response regions.

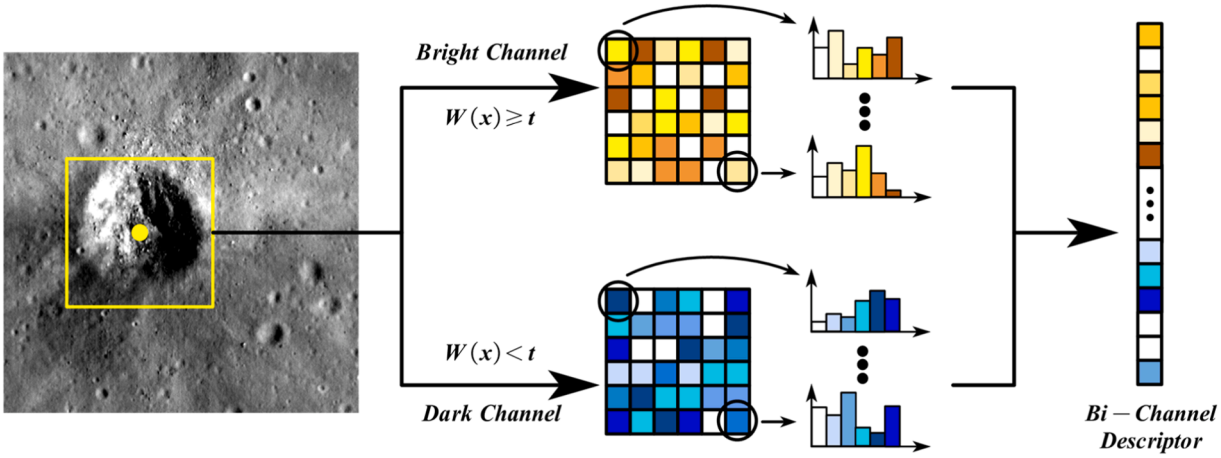

Fig. 4. Illustration of the bright-dark bi-channel descriptor construction pipeline.

To establish a comparable ranking criterion across the two channels, a unified score is constructed based on structural response and photometric reliability. The score of a candidate pixel x is defined as

$$S_c(x) = R_c(x)\, w_{soft}(x) \quad , \quad c \in \{M_{PW}, m_{PW}\} \tag{9}$$

where $R_c(x)$ denotes the structural response in channel c. This scoring rule prioritizes candidates with both high structural saliency and high photometric reliability. A pixel is included as a candidate only when it lies within the valid region, satisfies the local extremum condition in the corresponding channel, and passes the adaptive quality screening. Pixels are excluded when they belong to invalid regions, isolated background responses, low-score areas, or locations too close to stronger selected candidates. Therefore, the sampling process is explicitly constrained by structural response, photometric reliability, valid-region consistency, and spatial non-redundancy. Fig. 3 compares the keypoint distributions before and after introducing photometric weighting. Without photometric weighting, some keypoints are still selected in weak-texture or low-reliability regions. After weighting, the selected keypoints are more concentrated on photometrically reliable terrain structures, and their distribution on the photometric reliability map shows better consistency with the reliability prior.

Candidates from the two channels and all pyramid levels are merged into a unified candidate set. Each candidate retains its original-image coordinate, pyramid scale, channel label, and score. The candidate set is sorted in descending order by $S_c(x)$, and a minimum-distance constraint is enforced during selection to avoid repeated sampling on the same local structure. After candidates from different pyramid levels are mapped back to the original image coordinate system, cross-scale near-neighbor suppression is applied to remove redundant detections from adjacent scales. Because the M_{PW} channel typically produces stronger responses, a purely global ranking may suppress most candidates from m_{PW}. To preserve structural diversity, a minimum retention ratio ρ is introduced for the m_{PW} channel. Specifically, up to ρN keypoints are first selected from the m_{PW} candidates that satisfy the spatial distance constraint, and the remaining quota is filled using all candidates until the target number N is reached. This mechanism maintains the dominance of M_{PW}, while ensuring that m_{PW} contributes complementary keypoints in weak-texture and brightness-inversion regions.

For each selected keypoint, a dominant orientation is further estimated to support rotation-normalized descriptor construction. A local orientation histogram is built from the *MIM* map within a neighborhood centered at the keypoint. Each neighboring pixel votes to its corresponding *MIM* orientation bin, and the voting weight is determined by the local structural strength, photometric soft weight, and Gaussian spatial weight. For a keypoint p, the orientation histogram can be written as

$$H_p(k) = \sum_{x \in \mathcal{N}(p)} M_{PW}(x) w_{\text{soft}}(x) G_\sigma(x-p) 1[MIM(x) = k], \quad k = 0, \ldots, K-1 \tag{10}$$

where $N(p)$ denotes the local neighborhood, $G_\sigma(\cdot)$ is the Gaussian spatial weighting function, $1[\cdot]$ is the indicator function, and $K = 12$ is the number of orientation bins. The dominant orientation bin is determined by the maximum histogram response:

$$k_p = \text{argmax}_k H_p(k) \tag{11}$$

The selected orientation provides a canonical reference for local coordinate normalization. For a pixel with relative offset (dx, dy) around keypoint p, the normalized local coordinates are computed as

$$x_r = \cos\theta_p dx - \sin\theta_p dy, \quad y_r = \sin\theta_p dx + \cos\theta_p dy \tag{12}$$

where θ_p is the orientation angle corresponding to k_p. Meanwhile, the local *MIM* index is shifted to a relative orientation bin:

$$b_{\text{norm}}(x) = \left(MIM(x) - k_p + K\right) \bmod K \tag{13}$$

This rotation normalization allows the subsequent descriptor to encode local structures in a canonical orientation frame, thereby reducing the sensitivity of the descriptor to in-plane rotation.

The final keypoint set lies within the valid region, satisfies the adaptive quality and spatial distance constraints, and contains the position, scale, channel label, score, and dominant orientation of each keypoint. Combined with *MIM*,*W* and w_{soft}, this keypoint set provides a stable and controllable basis for the subsequent rotation-normalized bright-dark descriptor construction and geometric-consistency verification.

3.5. Rotation-Normalized bright-dark bi-channel descriptor and swap-aware matching

After obtaining the photometrically weighted PC representation and the rotation-normalized keypoint set, a local descriptor is constructed to improve cross-image comparability under strong illumination differences, brightness inversion, and in-plane rotation. In conventional phase-congruency-based descriptors, the local neighborhood of each keypoint is divided into spatial cells, and orientation histograms are accumulated using the dominant orientation labels provided by *MIM*. Although this representation is less sensitive to image magnitude than gradient-based descriptors, two types of degradation may still occur under complex illumination. First, pixels in shadow and penumbra regions have lower photometric reliability, and equal-weight histogram accumulation may introduce noise-driven distribution shifts. Second,

brightness inversion can transfer structural responses between bright and dark regions, causing a single-channel histogram to mix different irradiance states. To address these issues, PWIFT constructs a photometric-weighted rotation-normalized bright-dark bi-channel descriptor and further adopts a swap-aware distance for descriptor matching. Fig. 4 illustrates the construction pipeline of the bright-dark bi-channel descriptor. After local rotation normalization, pixels within each descriptor cell are divided into bright and dark channels according to the photometric reliability state, and the corresponding orientation histograms are accumulated and concatenated to form the final bright-dark bi-channel descriptor.

For each selected keypoint x_i, a fixed-size local patch is extracted from *MIM*, M_{PW}, and the photometric weight map W at the corresponding pyramid level. The patch is first normalized according to the dominant orientation θ_i obtained in Section 3.4. The normalized local coordinates and relative orientation bins are then used for descriptor construction. The patch is divided into $n_o \times n_o$ spatial cells, and each cell accumulates a histogram with n_{bins} orientation bins. For a pixel x in the local patch, its descriptor voting weight is defined as

$$w_i(x) = M_{PW}(x) w_{\text{soft}}(x) G_\sigma(x - x_i) \tag{14}$$

where $G_\sigma(\cdot)$ is the Gaussian spatial weighting function. This weighting rule assigns larger contributions to structurally salient and photometrically reliable pixels, while continuously downweighting low-reliability pixels rather than discarding them directly. For an arbitrary cell c, the rotation-normalized orientation histogram is written as

$$h_c(k) = \sum_{x \in c} w_i(x) 1[b_{\text{norm}}(x) = k], \quad k = 0, \ldots, n_{bin} - 1 \tag{15}$$

where $b_{norm}(x)$ is the relative *MIM* orientation bin after dominant-orientation normalization.

To reduce descriptor ambiguity caused by brightness inversion, pixels in each cell are further divided into bright and dark sets according to the photometric weight threshold $t = 0.2$. The two sets are accumulated into separate channel histograms:

$$h_c^{\text{Bright}}(k) = \sum_{x \in c,\ W(x) \geq t} w_i(x) 1[b_{\text{norm}}(x) = k] \tag{16}$$

$$h_c^{\text{Dark}}(k) = \sum_{x \in c,\ W(x) < t} w_i(x) 1[b_{\text{norm}}(x) = k] \tag{17}$$

For each cell, h_c^{Bright} and h_c^{Dark} are concatenated in a fixed order, and all cells are further concatenated to form the final descriptor D_i. With $n_o = 4$, $n_{bin} = 12$, and two irradiance channels, the descriptor dimensionality is

$$\dim(D_i) = n_o \times n_o \times n_{bin} \times 2 = 4 \times 4 \times 12 \times 2 = 384 \tag{18}$$

The resulting bright-dark bi-channel descriptor jointly encodes relative structural orientation and local irradiance state. When brightness swapping occurs between two images, structural responses can be compared through the bright and dark channels instead of being mixed into a single histogram. After concatenation, the descriptor is processed by L2 normalization, value clipping, and a second L2 normalization to reduce scale differences and suppress dominant histogram peaks.

In the matching stage, descriptors extracted from all pyramid levels are mapped back to the original image coordinate system and merged into a unified descriptor set for each image. To handle possible bright-dark exchange under illumination reversal, descriptor distance is computed in a swap-aware manner. Let $D_i = \left[D_i^{Bright}, D_i^{Dark}\right]$ and $D_j = \left[D_j^{Bright}, D_j^{Dark}\right]$ denote two descriptors. Their swap-aware distance is defined as

$$d_{\text{swap}}(D_i, D_j) = \min\left(d_{\text{ori}}(D_i, D_j), d_{\text{ex}}(D_i, D_j)\right) \tag{19}$$

where

$$d_{\text{ori}}(D_i, D_j) = \sqrt{\| D_i^{\text{Bright}} - D_j^{\text{Bright}} \|_2^2 + \| D_i^{\text{Dark}} - D_j^{\text{Dark}} \|_2^2} \tag{20}$$

$$d_{\text{ex}}(D_i, D_j) = \sqrt{\| D_i^{\text{Bright}} - D_j^{\text{Dark}} \|_2^2 + \| D_i^{\text{Dark}} - D_j^{\text{Bright}} \|_2^2} \tag{21}$$

Here, d_{ori} measures the distance under the original bright-dark order, whereas d_{ex} measures the distance after exchanging the two irradiance channels. The smaller distance is used for nearest-neighbor matching. The resulting bright-dark bi-channel descriptor jointly encodes relative structural orientation and local irradiance state. When brightness swapping occurs between two images, structural responses can be compared through the bright and dark channels instead of being mixed into a single histogram. After concatenation, the descriptor is processed by L2 normalization, value clipping, and a second L2 normalization to reduce scale differences and suppress dominant histogram peaks. Therefore, the descriptor provides a compact and illumination-aware representation for comparing local structures under illumination reversal, shadow migration, and in-plane rotation.

3.6. Coarse-to-Fine rotation-scale search and geometry-guided match refinement

Based on the PWIFT feature-construction modules described above, PWIFT performs a geometry-guided matching stage to handle large inter-image rotation-scale variations. Let I_r and I_m denote the reference image and the moving image to be registered, respectively. Instead of directly applying full matching to the original image pair, PWIFT first searches for a reliable rotation-scale hypothesis that transforms I_m into a candidate frame closer to I_r. The optimal candidate is selected according to the geometric consistency of tentative matches, including the inlier count, inlier ratio, and reprojection error of the estimated homography.

A set of scale candidates is first constructed as

$$S_{rs} = \{0.5, 0.707, 1.0, 1.414, 2.0, 3.0\} \tag{22}$$

For each scale candidate $s \in S_{rs}$, the moving image I_m is scaled into a candidate frame and evaluated using a lightweight matching stage. A lightweight evaluation is performed at a reduced image resolution with a limited number of keypoints and candidate matches. This stage provides a fast estimate of the geometric consistency of each hypothesis, while the final correspondences are produced in the full PWIFT matching stage. Each candidate is ranked using a lexicographic geometric-consistency criterion, where the number of geometric inliers is used as the primary criterion, followed by the inlier ratio, the reprojection RMSE of the estimated homography, and the number of raw matches. The two highest-ranked scale candidates are retained for the subsequent coarse rotation search. For each retained scale candidate, a coarse rotation search is performed over a full-range rotation set:

$$\Theta_c = \{-180^\circ, -120^\circ, -60^\circ, 0^\circ, 60^\circ, 120^\circ\} \tag{23}$$

Each rotation-scale candidate $C = (s, \theta)$ defines a transformation $T_{m \to c}(s, \theta)$, which maps the moving image to the candidate frame. Candidate matching is then performed between the reference image and the transformed moving image, and the same lexicographic geometric-consistency criterion is used for candidate ranking. The two highest-ranked rotation candidates are further refined locally. For a selected coarse angle θ, the local refinement offsets are defined as

$$\Delta\Theta = \{-30^\circ, -15^\circ, 0^\circ, 15^\circ, 30^\circ\} \tag{24}$$

The refined rotation candidates are therefore $\theta + \Delta\theta$, where $\Delta\theta \in \Delta\Theta$. All refined candidates are evaluated using the lightweight matching stage, and the final rotation-scale hypothesis is selected as

$$C^* = (s^*, \theta^*) = \text{argmax}_C \Gamma(C) \tag{25}$$

Table 1
Benchmark image groups and subset configuration.

Subset	Site	Image ID	Incidence angle (°)	Emission angle (°)	Phase angle (°)	GSD (m/px)	Pairs	Rotation/Scale
E010-0.54 m Illumination	NAC_PHO_E010N0230	M150368601 (original)	63.43	29.02	34.41	0.54	597	No
		M129133239R	53.00	6.00	48.00	0.54	597	
		M109080308R	2.00	16.00	16.00	0.54	597	
E010-1 m Control	NAC_PHO_E010N0230	M150368601 (resampled)	63.43	29.02	34.41	1.00	888/subset	Yes
		M1114021499	44.56	22.43	22.13	1.00		
		M1192915976	18.19	1.16	17.03	1.00		
E064-3 m Control	NAC_PHO_E064S3160	M1219245090	63.42	16.70	80.07	3.00	90/subset	Yes
		M1282753376	24.80	1.72	26.46	3.00		
		M1254537038	7.00	1.15	6.48	3.00		
E207-5 m Control	NAC_PHO_E207N3357	M1160269154	47.74	18.69	30.81	5.00	129/subset	Yes
		M183689789	40.70	1.19	39.58	5.00		
		M1239124122	21.58	21.18	9.45	5.00		

Note: "Pairs" denotes the number of final patch pairs after cropping, valid-region screening, and quality control. For the E010-0.54 m illumination subset, the three two-image illumination combinations contain 1791 pairs in total. For the controlled subsets, "pairs/subset" denotes the number of pairs in each of the illumination-only, rotation-only, scale-only, and rotation-scale subsets. "Rotation/Scale" indicates whether controlled rotation-only, scale-only, and rotation-scale subsets were constructed. "No" indicates illumination-only pairs without additional geometric perturbations, whereas "Yes" indicates that controlled geometric perturbations were additionally introduced.

where $\Gamma(C)$ denotes the lexicographic geometric-consistency ranking criterion. Through this coarse-to-fine pruning process, PWIFT first reduces the search space from all scale candidates to the most promising scale hypotheses, then narrows the full-range rotation search to locally refined rotation-scale candidates.

After obtaining C^*, the moving image is transformed by $T_{m\to c}(s^*, \theta^*)$ and the full PWIFT matching process is performed between the reference image and the transformed moving image. This full matching stage uses the complete photometric-weighted feature construction, keypoint detection, local rotation normalization, and rotation-normalized bright-dark descriptor described in Sections 3.2–3.5. After matching in the selected candidate frame, the moving-side keypoints are transformed back to the original coordinate system of I_m using the inverse candidate transformation $T_{m\to c}^{-1}(s^*, \theta^*)$. Therefore, all subsequent matching refinement and result reporting are performed in the original coordinate systems of I_r and I_m.

Under the selected rotation-scale hypothesis, descriptor matching is performed between the reference image and the transformed moving image using the swap-aware distance between the bright and dark channels. Specifically, both the original bright-dark channel order and the exchanged channel order are considered, and the smaller distance is used for nearest-neighbor matching. The obtained descriptor matches are then filtered by mutual check to remove many-to-one correspondences. FSC is subsequently performed under a homography model to remove geometrically inconsistent matches. The estimated homography from the reference image to the moving image is denoted as $\widehat{H}_{r\to m}$, and

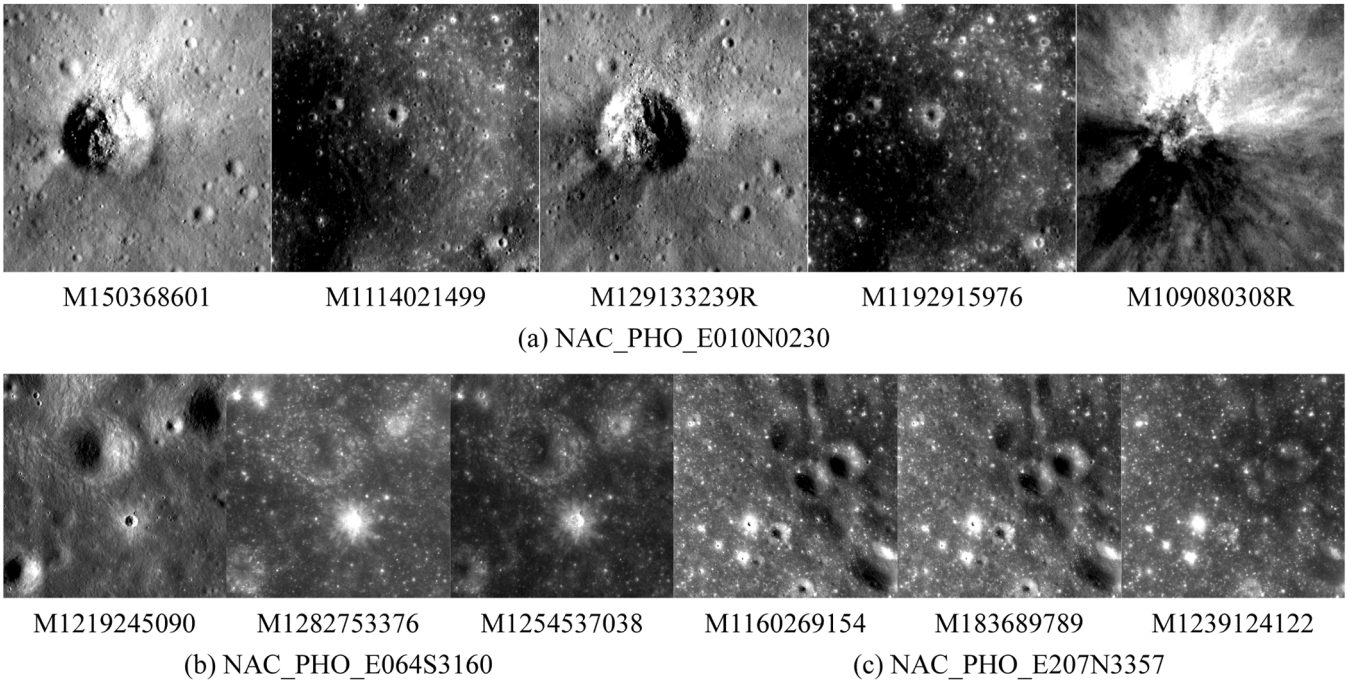

Fig. 5. Representative image patches from the lunar multi-illumination image registration benchmark. (a) NAC_PHO_E010N0230 with five illumination states, (b) NAC_PHO_E064S3160 with three illumination states, and (c) NAC_PHO_E207N3357 with three illumination states. The image IDs are shown below the corresponding patches.

the geometrically filtered match set is denoted as M_F.

To further suppress residual outliers after FSC filtering, a homography-based reprojection cleanup is applied. The geometrically filtered match set $M_F$is used to estimate $\widehat{H}_{r\to m}$. For each match $\left(p_{r,i}, p_{m,i}\right) \in M_F$, the estimated-H reprojection error is computed as

$$e_{\text{est},i} = \| p_{m,i} - \pi\left(\widehat{H}_{r\to m}\widetilde{p}_{r,i}\right) \|_2 \tag{26}$$

Only matches with reprojection error no larger than τ_e are retained:

$$M_{clean} = \left\{ \left(p_{r,i}, p_{m,i}\right) \in M_F | e_{\text{est},i} \le \tau_e \right\} \tag{27}$$

In this study, $\tau_e = 5$ pixels is used. The retained correspondences form the cleaned final match set. Through the combination of coarse-to-fine rotation-scale search, full PWIFT matching, FSC filtering, and estimated-H reprojection cleanup, PWIFT obtains more stable final correspondences under large rotation-scale variations and complex illumination differences.

4. Experimental details

4.1. Dataset and experimental settings

To systematically evaluate the robustness of lunar image matching algorithms under complex illumination and geometric perturbations, we constructed a lunar multi-illumination image registration benchmark. Three representative LROC NAC photometry sites were selected, including NAC_PHO_E010N0230, NAC_PHO_E064S3160, and NAC_PHO_E207N3357. These sites cover different lunar terrain structures, shadow distributions, and solar illumination geometries. Specifically, NAC_PHO_E010N0230 contains five illumination states, while NAC_PHO_E064S3160 and NAC_PHO_E207N3357 each contain three illumination states. In total, the benchmark includes 11 site-specific illumination states from three lunar regions. The illumination geometry, spatial resolution, subset configuration, pair counts, and rotation/scale perturbation settings are summarized in Table 1, and representative image patches are shown in Fig. 5. As illustrated in Fig. 5, the selected images exhibit pronounced appearance variations under different illumination geometries, including brightness inversion, shadow migration, and terrain-dependent contrast changes.

After image cropping, valid-region screening, and quality control, the final benchmark contains 6219 image pairs. As shown in Table 1, the benchmark images were organized into four image groups according to site, spatial resolution, and experimental purpose. For NAC_PHO_E010N0230, two groups were used. The E010-0.54 m Illumination subset consists of M150368601, M129133239R, and M109080308R at the original 0.54 m/px resolution. This subset contains 1791 illumination-only image pairs and was used for illumination-related analysis without additional rotation or scale perturbations. The E010-1 m Control subset consists of the resampled M150368601 image and the two 1 m/px images M1114021499 and M1192915976. This subset contains 888 image pairs in each of the illumination-only, rotation-only, scale-only, and rotation-scale subsets. Similarly, E064-3 m Control and E207-5 m Control contribute 90 and 129 image pairs to each corresponding subset, respectively.

Accordingly, the full illumination-only subset contains 2898 real multi-illumination image pairs, including 1791 pairs from E010-0.54 m Illumination, 888 pairs from E010-1 m Control, 90 pairs from E064-3 m Control, and 129 pairs from E207-5 m Control. To further evaluate geometric robustness, three controlled perturbation subsets, namely rotation-only, scale-only, and rotation-scale, were constructed. Each perturbation subset contains 1107 image pairs, including 888, 90, and 129 pairs from E010-1 m Control, E064-3 m Control, and E207-5 m Control, respectively. For the geometrically perturbed pairs, the relative rotation spans from -180° to 180°, and the relative scale ratio ranges from 0.5 to 3. The corresponding ground-truth homographies were recorded for all image pairs and used for quantitative matching evaluation.

To evaluate the matching performance of PWIFT under complex lunar illumination conditions, nine representative methods were selected for comparison, including five conventional feature-based methods, RIFT [40], WSSF [41], OFLP [20], GLIFT [21], and MISR [23], and four learning-based methods, SuperPoint [27] + LightGlue [28], LoFTR [29], XoFTR [30], and RoMaV2 [33]. All methods were tested on the original 512 × 512 image patches, and all output correspondences were represented in the original image coordinate system. RIFT, WSSF, OFLP, GLIFT, MISR and PWIFT retained their original feature extraction, descriptor construction, and initial matching procedures, while SuperPoint + LightGlue, LoFTR, XoFTR, and RoMaV2 were evaluated using publicly available pretrained models. The learning-based methods were evaluated using their publicly available pretrained models without fine-tuning on the lunar benchmark. Therefore, their results mainly reflect cross-domain generalization under planetary illumination and terrain conditions. To ensure a consistent post-processing condition, FSC was adopted for final mismatch removal in all methods, with the geometric consistency threshold set to 5 pixels. For PWIFT, the scale candidates were set to {0.5, 0.707, 1.0, 1.414, 2.0, 3.0}. The coarse rotation search covered -180° to 120° with a 60° interval, and local rotation refinement was performed using offsets of $\{-30^\circ, -15^\circ, 0^\circ, 15^\circ, 30^\circ\}$. The final rotation-scale candidate was determined according to the number of FSC inliers, the inlier ratio, and the reprojection error of the estimated transformation.

In the experiments, matching performance was evaluated using six groups of metrics: success rate (SR), mean error (ME@10), number of correct matches under a 5-pixel threshold (NCM@5), ratio of correct matches under a 5-pixel threshold (RCM@5), root mean square matching error ($\text{RMSE}_{\text{match}}$), and high-precision hit rate under a 1-pixel threshold (Hit@1). Let the FSC-filtered matching result of the $j-th$ image pair be:

$$M_j = \{(p_i, q_i)\}_{i=1}^{n_j} \tag{28}$$

where p_i and q_i denote the $i-th$ matched points in the reference and moving images, respectively, and n_j is the number of valid matched point pairs for this image pair. According to the geometric relationship recorded in the dataset, p_i is projected onto the moving image coordinate system, and the error of the $i-th$ match is defined as:

$$e_i = \| \widehat{p}_i - q_i \|_2 \tag{29}$$

where $\widehat{p}_i$ denotes the projected position of p_i in the moving image coordinate system. The ME@10 and $\text{RMSE}_{\text{match}}$ of the $j-th$ image pair are computed as:

$$ME@10 = \frac{1}{NCM@10} \sum_{i=1}^{NCM@10} e_i \tag{30}$$

$$RMSE_{\text{match},j} = \sqrt{\frac{1}{n_j}\sum_{i=1}^{n_j} e_i^2} \tag{31}$$

where NCM@10 is defined as the number of correct matches with an error not larger than 10 pixels. RCM@5 is defined as the ratio of correct matches to the number of valid matches retained after FSC filtering. An image pair is considered successfully matched when $ME_j \le 5$ pixels. SR is defined as the ratio of successfully matched image pairs to all evaluated image pairs:

$$SR = \frac{N_{\text{success}}}{N_{\text{total}}} \tag{32}$$

Hit@1 is defined as the proportion of high-precision matches with an error not larger than 1 pixel. SR reflects image-pair-level matching stability, ME and $\text{RMSE}_{\text{match}}$ characterize geometric accuracy and

Table 2
Comparison of matching results using photometric reliability maps constructed by different photometric models.

Photometric Model	SR@5	ME@10	NCM@5	$RMSE_{match}$
KS	90.8%	3.6	96.5	4.1
LS-phase	90.8%	3.6	98.1	3.9
Minnaert	79.3%	3.8	71.8	4.3
Akimov	**94.1%**	**3.1**	**150.8**	**3.6**

residual error distribution, NCM@5 and RCM@5 measure the number and reliability of correct correspondences, and Hit@1 evaluates high-precision matching under a strict pixel-level tolerance. These metrics jointly provide the quantitative basis for the subsequent experimental analyses. All experiments are conducted on a workstation equipped with an Intel(R) Core(TM) Ultra 9 275HX CPU, 32 GB RAM, and an NVIDIA GeForce RTX 5060 Laptop GPU.

4.2. Selection of the photometric model

To achieve more stable feature repeatability under extreme illumination, this section compares four candidate models for constructing the photometric weight map. The comparison was conducted using the M109080308R–M129133239R image-pair group, which was selected from the E010-0.54 m Illumination subset and contains 597 illumination-only patch pairs without additional rotation or scale perturbations. The candidates include the Kaasalainen-Shkuratov photometric model (KS) [42], the Lommel-Seeliger disk function with a phase function (LS-phase) [43], the Minnaert photometric function [44], and the Akimov disk function [45]. The objective is to construct pixel-level weights that can suppress unstable radiometric and brightness variations caused by shadows and brightness inversion, thereby improving matching robustness and reducing geometric errors.

The Kaasalainen-Shkuratov model uses a separable form with a phase function and a disk function. The phase term characterizes global brightness attenuation with phase angle, while the disk function describes disk-wide brightness distribution. It has been used in photometric processing and comparative analysis of planetary mission data. The LS-phase model uses the Lommel-Seeliger disk function to describe brightness variations dominated by single scattering, and introduces a phase term to compensate for global darkening caused by phase-angle changes. The Minnaert function adjusts the strength of disk darkening through an exponential parameter. Although it originates from classical lunar photometry studies, its parameters are sensitive to surface properties and imaging geometry, which may reduce cross-scene stability. The Akimov disk function was developed from lunar photometric observations and has been widely adopted for disk-function fitting and photometric correction of the Moon and small bodies. It provides broad geometry applicability and strong empirical robustness. Matching results using photometric reliability maps constructed by these four models are reported in Table 2.

As shown in Table 2, the Akimov disk function achieves the best overall performance among the four candidates, with the highest SR@5 of 94.1%, the largest NCM@5 of 150.8, the lowest $RMSE_{match}$ of 3.6, and the lowest ME@10 of 3.1. Compared with the KS and LS-phase models, the Akimov model not only improves the success rate, but also substantially increases the number of correct matches while reducing geometric errors. The Minnaert function shows a clear degradation in both SR and NCM, indicating lower stability under complex illumination. Considering its superior matching robustness, lower geometric error, and mature application in lunar photometry and disk-function modeling, the Akimov disk function is selected as the photometric weighting model for PWIFT in the subsequent experiments.

4.3. Comparative evaluation under illumination-only variations

(a) Overall quantitative comparison

Table 3 summarizes the overall quantitative results of the ten methods on the full illumination-only subset with 2898 image pairs. This subset includes E010-0.54 m Illumination, the illumination-only part of E010-1 m Control, E064-3 m Control, and E207-5 m Control. The compared metrics include SR@5, ME@10, NCM@5, RCM@5, $RMSE_{match}$, and Hit@1, which jointly evaluate image-pair-level matching stability, geometric accuracy, matching density, correspondence reliability, and high-precision matching ability.

As shown in Table 3, PWIFT achieves the highest success rate among all methods, with an SR@5 of 97.93%. This result indicates that PWIFT provides the most stable registration performance under real lunar illumination variations. In terms of geometric accuracy, PWIFT obtains an ME@10 of 2.66 pixels and an $RMSE_{match}$ of 2.67 pixels, showing that its high success rate is achieved while maintaining a low matching error. In addition, PWIFT reaches an RCM@5 of 90.80%, which is the second highest among all methods and is close to the best value obtained by XoFTR. This demonstrates that most correspondences retained by PWIFT after FSC filtering are geometrically reliable.

Among the learning-based methods, SuperPoint + LightGlue achieves a low ME@10 of 2.20 pixels, but its SR@5 and NCM@5 are limited, reflecting insufficient repeatability and match density in lunar scenes. XoFTR achieves the best ME@10, RCM@5, and Hit@1 values, indicating strong local matching accuracy and high-precision correspondence quality. However, its SR@5 of 95.51% is lower than that of PWIFT. LoFTR and RoMaV2 produce a larger number of matches, with NCM@5 values of 1042.59 and 1507.87, respectively, but their success rates decrease to 80.33% and 78.85%. This suggests that dense or semi-dense learning-based matchers can provide abundant correspondences, but their performance is less stable under large illumination differences and lunar-domain appearance variations.

For conventional methods, WSSF obtains a high SR@5 of 94.82% and a large NCM@5 of 689.69, showing better robustness than RIFT and OFLP under illumination changes. However, its ME@10 and Hit@1 are inferior to those of PWIFT and stronger learning-based methods. RIFT and OFLP produce considerable numbers of correct matches, but their SR@5 values are 83.13% and 86.82%, respectively, indicating that phase-congruency-based or local structural descriptors alone are

Table 3
Overall quantitative comparison of different methods on the illumination-only subset.

Algorithm	SR@5/%	ME@10	NCM@5	RCM@5/%	$RMSE_{match}$	Hit@1/%
PWIFT	**97.93**	2.66	403.08	90.80	2.67	5.21
RIFT	83.13	2.79	863.67	79.24	2.51	1.93
WSSF	94.82	3.05	689.69	81.70	2.67	1.41
OFLP	86.82	2.84	826.44	82.84	2.48	1.55
GLIFT	75.16	1.99	508.15	69.37	1.82	28.62
MISR	26.33	1.95	27.58	27.89	1.45	12.53
SuperPoint+LightGlue	78.47	2.20	352.42	76.16	2.24	5.00
LoFTR	80.33	2.41	1042.59	80.62	**1.42**	18.98
XoFTR	95.51	**1.94**	1005.86	**91.88**	1.78	**30.40**
RoMaV2	78.85	2.44	**1507.87**	80.70	2.05	22.53

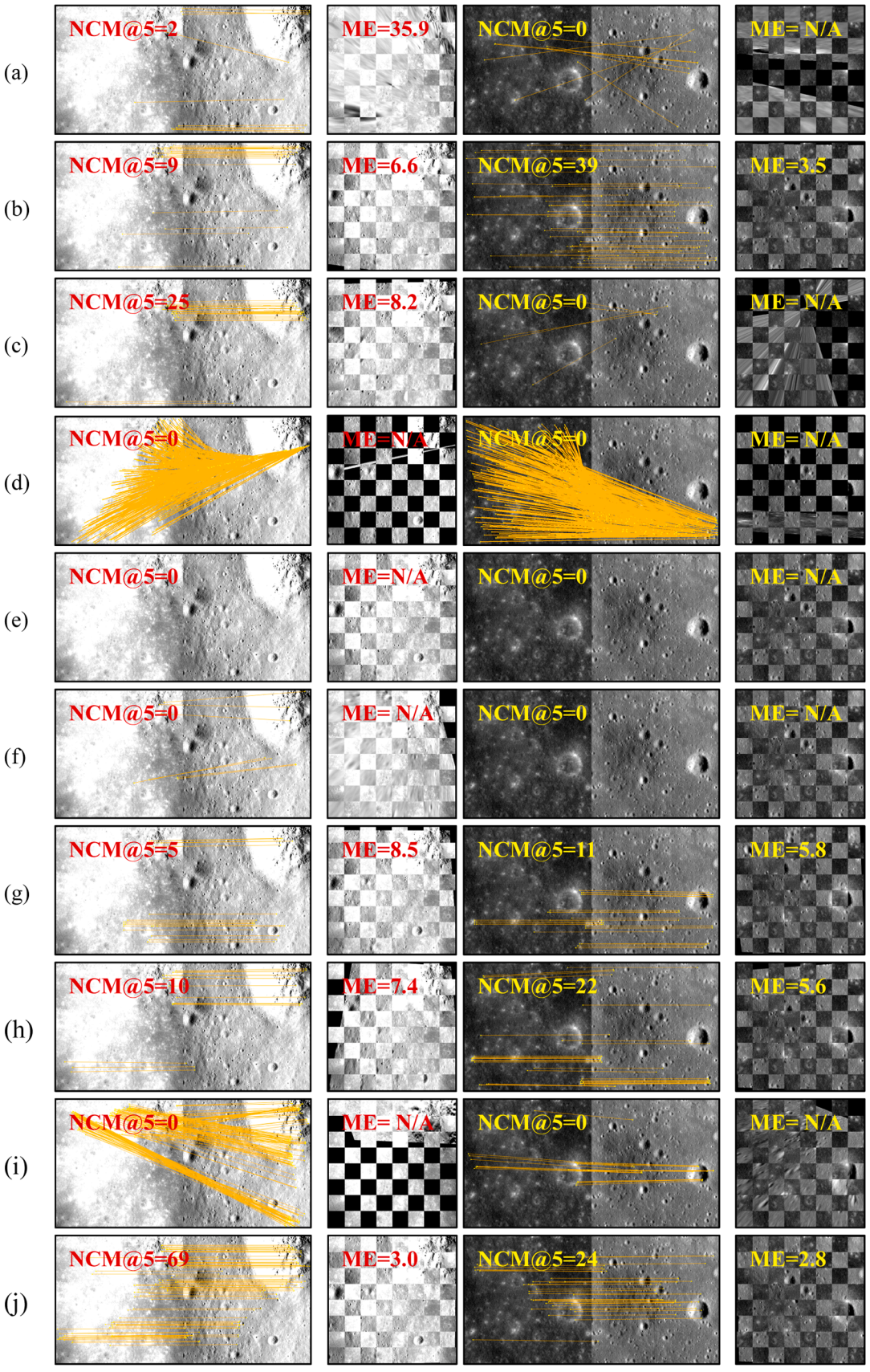

Fig. 6. Qualitative comparison of matching results on two representative illumination-only image pairs. (a) RIFT, (b) WSSF, (c) OFLP, (d) GLIFT, (e) MISR, (f) SuperPoint + LightGlue, (g) LoFTR, (h) XoFTR, (i) RoMaV2, (j) PWIFT.

insufficient for stable registration under complex lunar illumination. GLIFT achieves a relatively low ME@10 of 1.99 pixels and a high Hit@1 of 28.62%, suggesting that it can obtain accurate correspondences when reliable matches are established. However, its SR@5 decreases to 75.16%, and its RCM@5 is 69.37%, indicating reduced image-pair-level stability on the full illumination-only subset. MISR obtains a low ME@10 of 1.95 pixels, but its SR@5, NCM@5, and RCM@5 are only 26.33%, 27.58, and 27.89%, respectively. This result shows that low matching error on successful pairs does not necessarily lead to stable registration across all complex illumination cases.

Overall, PWIFT provides the most balanced performance among compared methods. Although XoFTR achieves higher local precision and RoMaV2 generates more correct matches, PWIFT achieves the best registration stability while maintaining competitive geometric accuracy and correspondence reliability. Compared with recent feature-based methods such as OFLP and GLIFT, PWIFT is more robust at the image-pair level under severe lunar illumination variations, because its photometric reliability prior is propagated through structural feature construction, keypoint selection, descriptor voting, and geometric refinement. This advantage stems from PWIFT's photometric-aware design. The photometric weight map suppresses unreliable structures in deep-shadow and strongly illuminated regions, improving keypoint repeatability under nonuniform illumination. The bright-dark bi-channel descriptor reduces descriptor confusion caused by brightness inversion and shadow migration. Additionally, the rotation-scale search improves PWIFT's tolerance to residual geometric differences between illumination-varying image pairs. These designs enable PWIFT to retain sufficient reliable correspondences while avoiding the instability of conventional structural descriptors and domain sensitivity of learning-based matchers.

(b) Qualitative comparison

To complement the quantitative evaluation, qualitative comparisons were conducted on two representative image pairs selected from the full illumination-only subset with distinct illumination challenges. The first pair corresponds to a high-illumination case with weakened texture contrast, while the second pair represents a medium-illumination case with a large incidence-angle difference. Fig. 6 presents the qualitative comparison of the ten methods on two representative illumination-only image pairs. Rows (a)-(j) correspond to RIFT, WSSF, OFLP, GLIFT, MISR, SuperPoint + LightGlue, LoFTR, XoFTR, RoMaV2, and PWIFT, respectively. For each method, the FSC-retained match connections and the corresponding checkerboard registration results are shown. The left example represents a high-illumination case with noticeable illumination differences, where strong illumination weakens local texture contrast and reduces the number of repeatable features. The right example represents a medium-illumination case with a large incidence-angle difference, where local structures remain visible but the appearance of corresponding features changes significantly, increasing the risk of mismatches.

In the high-illumination example, most methods suffer from either insufficient correct matches or large registration errors. RIFT retains only two correct matches and produces a large ME of 35.9 pixels, indicating that the phase-congruency-based descriptor becomes unstable when strong illumination weakens local texture contrast. GLIFT generates many match connections in the visualization, but none of them satisfy the NCM@5 criterion, and its ME is therefore unavailable. This indicates that GLIFT can still produce abundant tentative correspondences in this case, but the retained matches are geometrically inconsistent under the strong illumination difference. MISR does not obtain valid correct correspondences in this example, with NCM@5 equal to 0 and ME unavailable. SuperPoint + LightGlue and RoMaV2 fail to obtain reliable correct correspondences, while WSSF, OFLP, LoFTR, and XoFTR retain only a limited number of correct matches and their ME values remain larger than 5 pixels. In contrast, PWIFT achieves the largest NCM@5 of 69 and the lowest ME of 3.0 pixels in this example. The match-connection visualization shows that PWIFT retains more spatially distributed correspondences across the illuminated and weak-texture regions, rather than concentrating only on a few high-contrast structures. The checkerboard result also shows more continuous crater rims and terrain boundaries across block interfaces, indicating more accurate geometric alignment. This improvement benefits from the photometric weighting strategy, which suppresses unstable responses in over-bright and low-reliability regions, and from the bright-dark bi-channel descriptor, which improves descriptor consistency under illumination-induced contrast changes.

In the medium-illumination example with a larger incidence-angle difference, the performance gap among the methods becomes more evident. WSSF produces a relatively uniform distribution of correct matches and achieves a valid registration result, but its ME is 3.5 pixels. GLIFT again produces many visually apparent match connections, but its NCM@5 remains 0 and ME is unavailable, suggesting that the correspondences are dominated by incorrect geometric associations rather than reliable homologous points. MISR also fails to generate correct matches in this example. PWIFT achieves a lower ME of 2.8 pixels while retaining more correct matches, with NCM@5 increasing to 24. The matching lines of PWIFT are distributed across a broader range of structural regions rather than being concentrated in a small local area. This spatially balanced distribution provides more stable constraints for the estimated transformation. The checkerboard image further confirms this advantage: crater boundaries and small terrain details are more continuously connected at the block interfaces, whereas methods with fewer or less reliable matches show visible block discontinuities or local structural offsets.

The visual differences in Fig. 6 are consistent with the design of PWIFT. The photometric weight map suppresses unreliable responses in over-bright, shadowed, or low-reliability regions, reducing the influence of unstable structures on keypoint detection and descriptor construction. The bright-dark bi-channel descriptor further alleviates descriptor ambiguity caused by brightness inversion and shadow migration. As a result, PWIFT does not simply maximize the number of matches; instead, it tends to retain correspondences with higher geometric reliability and better spatial coverage. This explains why PWIFT can achieve lower ME even when its NCM@5 is slightly lower than some dense or semi-dense matching methods. Overall, the qualitative results show that PWIFT provides a more reliable balance between match quantity, spatial distribution, and geometric accuracy under illumination-only lunar image matching.

(c) Terrain-dependent performance analysis

To examine the influence of local terrain characteristics on matching performance, the illumination-only version of the controlled subsets was used for terrain-dependent analysis. This dataset contains 1107 image pairs, including 888 pairs from E010-1 m Control, 90 pairs from E064-3 m Control, and 129 pairs from E207-5 m Control, without additional rotation or scale perturbations. Specifically, the robust elevation range was defined as the difference between the 95th and 5th percentiles of valid DTM elevation values. Based on the global tertiles of this descriptor, all image pairs were divided into three terrain-relief groups: Low-relief, Medium-relief, and High-relief, with 369 pairs in each group.

As shown in Table 4, PWIFT achieves SR@5 values of 100.00% and 99.19% in the Low-relief and Medium-relief groups, respectively, while the SR@5 decreases to 92.68% in the High-relief group. This suggests that larger terrain relief may increase the probability of registration failure under varying illumination. However, the High-relief group also obtains the lowest ME@10 of 1.84 px. This result indicates that relief-rich regions may provide more distinctive structures, such as crater rims, slope boundaries, and shadow edges, which can improve local matching accuracy when registration succeeds. Therefore, terrain relief has a dual effect: it may increase the risk of failure in locally complex

Table 5
Site-level matching performance and terrain relief statistics.

Site	Pairs	Elev. range P95–P5 (m)	SR@5	ME@10	NCM@5	RCM@5
E010N0230	888	11.99	100.00	2.06	303.17	97.97
E064S3160	90	25.87	66.67	1.67	219.09	68.52
E207N3357	129	25.56	100.00	1.31	813.72	99.86

Table 4
Matching performance of PWIFT under different terrain relief levels.

Relief group	Pairs	Elev. range P95–P5 (m)	SR@5	ME@10	NCM@5	RCM@5
Low-relief	369	7.06	100.00	2.09	304.74	97.87
Medium-relief	369	11.20	99.19	1.91	387.67	97.65
High-relief	369	25.84	92.68	1.84	375.07	91.87

areas, while also providing stronger structural constraints for successful matches.

The site-level results in Table 5 further show that the effect of terrain relief is site-dependent. E064S3160 and E207N3357 have similar average elevation ranges, 25.87 m and 25.56 m, respectively, but their matching results differ markedly. E207N3357 achieves 100.00% SR@5, 813.72 NCM@5, and 99.86% RCM@5, whereas E064S3160 obtains only 66.67% SR@5, 219.09 NCM@5, and 68.52% RCM@5. This comparison indicates that terrain relief alone cannot fully explain the performance differences.

The illumination geometry provides a plausible explanation for this discrepancy. As shown in Table 1, E064S3160 has a larger incidence-angle range and phase-angle range than E207N3357. In particular, the low-sun image of E064S3160 has a phase angle of 80.07°, which may produce stronger shadow displacement and local radiometric changes. Under such conditions, terrain relief and illumination geometry jointly affect the repeatability of terrain structures. Although the photometric weighting in PWIFT suppresses illumination-sensitive regions, strong shadow and texture changes may still reduce the number and reliability of usable correspondences, as reflected by the lower NCM@5 and RCM@5 on E064S3160.

Overall, the terrain-dependent analysis shows that site-dependent performance differences are not determined by terrain relief alone. They are more likely related to the interaction among terrain relief, illumination geometry, shadow distribution, and local texture repeatability. This analysis provides a more specific explanation of the observed performance differences and supports the need for illumination-aware and geometry-guided matching in lunar image registration.

(d) Illumination-dependent performance analysis

To further analyze the sensitivity of different methods to illumination changes, the full illumination-only subset with 2898 image pairs was used, including 1791 pairs from E010-0.54 m Illumination, 888 pairs from E010-1 m Control, 90 pairs from E064-3 m Control, and 129 pairs from E207-5 m Control. This subset was divided into three illumination levels according to the absolute incidence angle. Images with incidence angles of 90°−60°, 60°−30°, and 30°−0° were assigned to the Low, Medium, and High illumination groups, respectively, corresponding to low-, medium-, and high-solar-elevation conditions. Based on these groups, three illumination combinations were evaluated: Low–Medium, Low–High, and Medium–High. The illumination range was computed as the difference between the maximum and minimum SR@5 values among the three combinations. A smaller illumination range indicates lower sensitivity to illumination-pair variations.

As shown in Table 6, most methods achieve high SR@5 values in the Low–Medium group, indicating that moderate illumination differences can be handled by both conventional and learning-based matchers. Except for MISR and LoFTR, all methods obtain SR@5 values above 90% in this group, and PWIFT achieves a high SR@5 of 99.90%. The Low–High group represents a near-terminator-like low-sun-angle illumination contrast, where the difference in shadow length, illumination direction, and bright-dark structure is more pronounced. In this group, WSSF and XoFTR achieve the highest SR@5 values of 98.96% and 98.86%, respectively. PWIFT obtains an SR@5 of 95.96%, which is slightly lower than the two best-performing methods but still remains at a high level, showing stable matching capability under strong illumination contrast.

The Medium–High group shows the clearest performance differences. PWIFT achieves the highest SR@5 of 97.93%, outperforming XoFTR, WSSF, OFLP, GLIFT, MISR, LoFTR, RIFT, RoMaV2, and SuperPoint + LightGlue. Several learning-based and conventional methods degrade in this group. This is consistent with lunar illumination characteristics: under higher solar-elevation conditions, cast shadows decrease and albedo-related brightness variations become more prominent, while under lower solar-elevation conditions, long shadows enhance local topographic structures. Therefore, a change from medium- to high-solar-elevation illumination alters the balance between shadow-defined structures and albedo-dominated texture patterns. This affects learning-based matchers relying on local appearance representations and conventional methods depending on local gradients or structural descriptors.

The illumination range further highlights the stability of PWIFT. PWIFT has the smallest illumination range of 3.94, which is substantially lower than XoFTR, WSSF, and the other comparison methods. This indicates that PWIFT is not necessarily the best method for every single illumination pair, but it exhibits the lowest performance fluctuation across different illumination combinations. The result suggests that the photometric weighting and bright-dark structural description in PWIFT help reduce illumination sensitivity and improve the consistency of matching performance under varying lunar illumination conditions.

(e) Matching density–accuracy trade-off

Table 6
SR@5 comparison under different illumination combinations.

Algorithm	Low–Medium	Low–High	Medium–High	Illumination range ↓
PWIFT	99.90	95.96	97.93	**3.94**
RIFT	99.90	94.82	54.66	45.24
WSSF	100.00	98.96	85.51	14.49
OFLP	99.90	85.71	74.84	25.06
GLIFT	100.00	86.15	39.94	60.06
MISR	33.15	1.26	39.02	37.76
SuperPoint+LightGlue	99.90	93.79	41.72	58.18
LoFTR	86.85	96.48	57.66	38.82
XoFTR	100.00	98.86	87.68	12.32
RoMaV2	91.72	96.48	48.34	48.14

Table 7
Quantitative statistics of the matching density–accuracy trade-off.

Method	SR@5	ME	Mean correct density ↑	Mean correct grid coverage ↑
PWIFT	98.32	3.21	15.82	0.551
RIFT	73.48	21.59	21.61	0.493
WSSF	91.90	3.61	14.93	0.557
OFLP	78.84	8.33	12.77	0.367
GLIFT	50.75	49.53	3.84	0.142
MISR	0.00	N/A	0.00	0.000
SuperPoint+LightGlue	75.11	30.78	7.91	0.397
LoFTR	71.25	6.22	18.99	0.392
XoFTR	93.08	3.11	27.95	0.413
RoMaV2	67.06	19.89	47.14	0.655

To further examine the relationship among match density, spatial distribution, and geometric accuracy, the E010-0.54 m Illumination subset was used. This subset contains 1791 illumination-only image pairs formed by M150368601, M129133239R, and M109080308R at NAC_PHO_E010N0230, without additional rotation or scale perturbations. For all methods, only the final matches retained by FSC were considered. Since no additional rotation or scale perturbation is introduced in this experiment, the direct pixel distance of the final matches was used to compute the mean matching error, denoted as ME, and ME = 5 px was used as the reference threshold. The correct match density was defined as NCM@5 normalized by the image area, while the spatial coverage was measured by the ratio of occupied cells in a 16×16 grid on the reference image. MISR is retained in Table 7 for completeness. Because it obtains an SR@5 of 0.00% on this subset, its ME is unavailable, while its correct match density and correct grid coverage are recorded as 0.

Fig. 7(a) shows the relationship between correct match density and ME. PWIFT samples are mainly concentrated in the lower-right region, indicating that PWIFT maintains low ME while preserving a relatively high correct match density. RoMaV2 achieves high correct match density for some image pairs, but many samples are still distributed in high-ME regions, indicating larger error fluctuations across image pairs. XoFTR also has many samples in the low-ME region, but its correct match density varies over a wider range. Fig. 7(b) shows a similar trend from the perspective of spatial coverage. PWIFT maintains low ME over a relatively wide range of occupied grid ratios, whereas RoMaV2 provides high spatial coverage but contains more high-error samples. This indicates that wider spatial coverage is meaningful only when the retained correspondences are also geometrically reliable. Fig. 7(c) and Table 7 summarize the method-level statistics. XoFTR obtains the lowest mean ME of 3.11 px, but its mean correct grid coverage is 0.413. RoMaV2 achieves the highest mean correct density and mean correct grid coverage, reaching 47.14 matches/10k pixels and 0.655, respectively, but its mean ME is 19.89 px and its SR@5 is 67.06%. In comparison, PWIFT achieves an SR@5 of 98.32%, a mean ME of 3.21 px, a mean correct density of 15.82 matches/10k pixels, and a mean correct grid coverage of 0.551, showing a stable density–accuracy balance.

Overall, higher match density or wider spatial coverage does not necessarily correspond to lower geometric error. PWIFT does not produce the largest number of matches, but its final matches show a more balanced behavior in terms of correct match density, spatial distribution, and ME control. This suggests that photometric-weight guidance and the bright–dark structural description help preserve spatially distributed reliable correspondences under varying illumination conditions. Therefore, PWIFT achieves a more robust density–accuracy trade-off by retaining sufficiently distributed and geometrically reliable matches, rather than maximizing match quantity alone.

4.4. *Robustness analysis of PWIFT under controlled geometric perturbations*

To evaluate the robustness of PWIFT under controlled geometric perturbations, four benchmark subsets were analyzed: illumination-only, rotation-only, scale-only, and rotation-scale. The illumination-only subset contains 2898 real multi-illumination image pairs and serves as the reference condition without additional geometric perturbations. The other three controlled perturbation subsets each contain 1107 image pairs, constructed from E010-1 m Control, E064-3 m Control, and E207-5 m Control using the same patch basis. These subsets

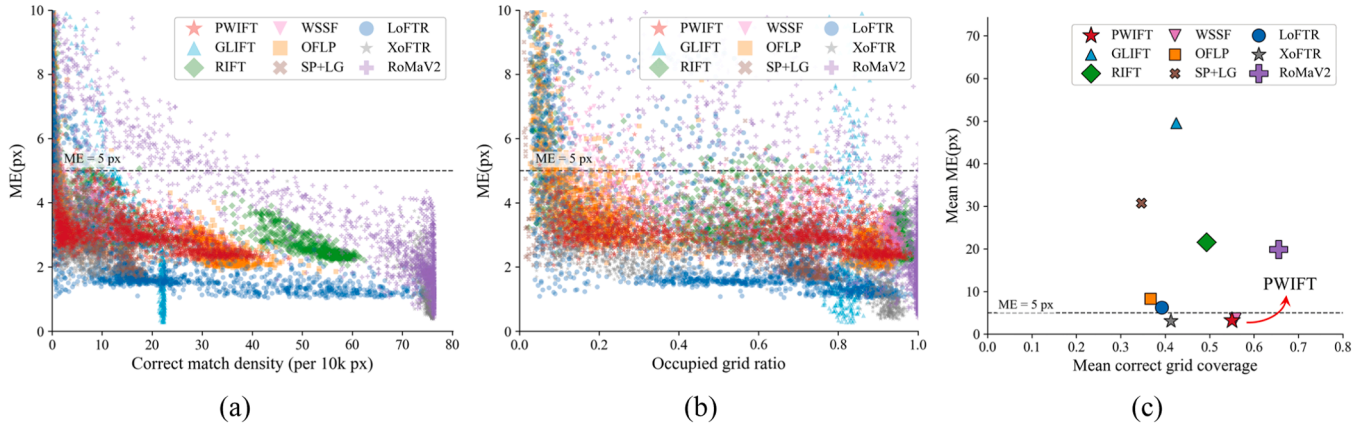

Fig. 7. Matching density–accuracy trade-off analysis. (a) Relationship between correct match density and ME. The correct match density is defined as NCM@5 normalized by the image area. (b) Relationship between spatial coverage and ME. The occupied grid ratio is computed using a 16×16 grid on the reference image based on the final matches. (c) Method-level statistics. The horizontal axis represents the mean correct grid coverage, the vertical axis represents the mean ME.

Table 8
Quantitative robustness of PWIFT under illumination and controlled geometric perturbations.

Subset	SR@5	ME@10	NCM@5	RCM@5	$RMSE_{match}$	Hit@1
Illumination-only	97.93	2.66	403.08	90.80	2.67	5.21
Rotation-only	95.21	2.09	251.24	93.53	2.28	0.09
Scale-only	88.89	2.04	176.39	86.80	2.19	1.36
Rotation-scale	80.13	2.07	152.23	80.40	2.23	0.72

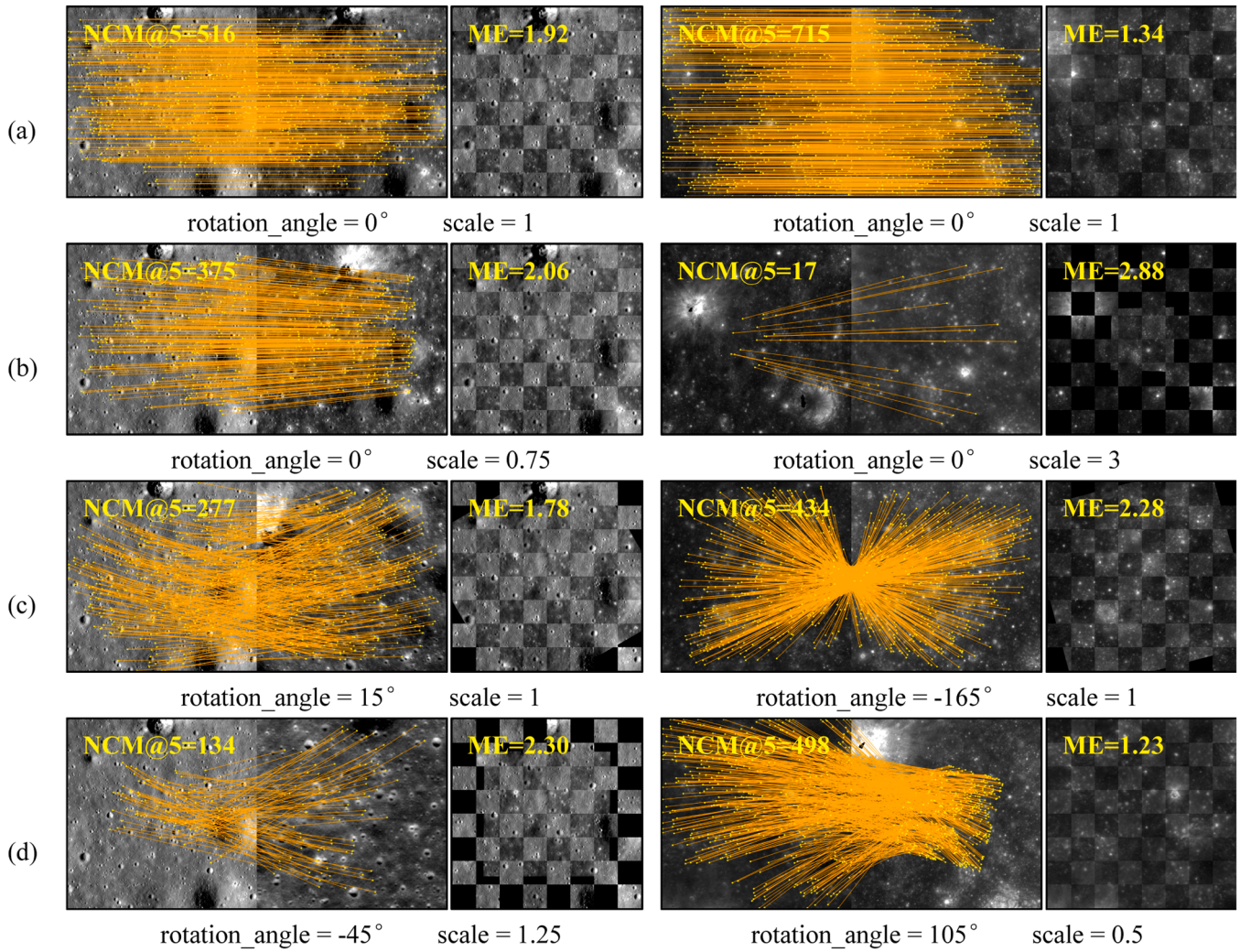

Fig. 8. Representative PWIFT registration results under controlled perturbations: (a) illumination-only, (b) scale-only, (c) rotation-only, and (d) rotation-scale.

were used to examine the sensitivity of PWIFT to rotation, scale, and their coupled variations. The quantitative results are summarized in Table 8.

PWIFT achieves an SR@5 of 97.93% on the illumination-only subset, with an ME@10 of 2.66 pixels and an RCM@5 of 90.80%, providing a stable baseline under photometric variations. When rotation perturbations are introduced, SR@5 remains high at 95.21%, and ME@10 decreases to 2.09 pixels, indicating that local rotation normalization and coarse-to-fine rotation search effectively reduce sensitivity to in-plane rotation. Under scale-only perturbations, SR@5 decreases to 88.89%, and NCM@5 decreases from 403.08 to 176.39, showing that scale variation reduces the number of repeatable correspondences. Nevertheless, ME@10 remains 2.04 pixels and RCM@5 remains 86.80%, suggesting that the retained matches are still geometrically reliable. The rotation-scale subset is the most challenging case, where PWIFT obtains an SR@5 of 80.13%, an ME@10 of 2.07 pixels, and an RCM@5 of 80.40%. Although coupled perturbations reduce the matching density and success rate, the registration error remains close to that of the rotation-only and scale-only subsets, indicating that the coarse-to-fine rotation-scale search and FSC-based filtering can preserve reliable correspondences under complex geometric changes.

Representative qualitative results are shown in Fig. 8. The illumination-only examples present dense and spatially consistent matches; the scale-only examples show fewer correct matches under large scale changes but still maintain well-aligned checkerboard results; the rotation-only examples confirm the ability of PWIFT to handle both moderate and large rotations; and the rotation-scale examples demonstrate that FSC-retained matches remain geometrically consistent even under coupled transformations.

Overall, Table 8 and Fig. 8 show that PWIFT maintains stable registration accuracy under controlled rotation-scale perturbations. As the geometric transformation becomes more complex, the matching density and success rate decrease, especially in the scale-only and rotation-scale subsets. However, ME@10 remains close to 2 pixels across the three geometric perturbation subsets, indicating that PWIFT preserves reliable correspondences through photometric-reliability-guided feature construction, rotation-normalized bright-dark description, coarse-to-fine rotation-scale search, and FSC-based verification.

4.5. Ablation study

The ablation study was conducted on the E010-0.54 m Illumination subset with 1791 image pairs. This subset consists of M150368601, M129133239R, and M109080308R at NAC_PHO_E010N0230 and contains only real illumination differences without additional rotation or scale perturbations. It was used to isolate the contributions of photometric reliability weighting, weight-guided keypoint selection, and the bright-dark bi-channel descriptor under strong illumination variations. Five variants were compared: RIFT, V1, V2, V3, and the full PWIFT. RIFT denotes the original baseline without photometric weighting. V1

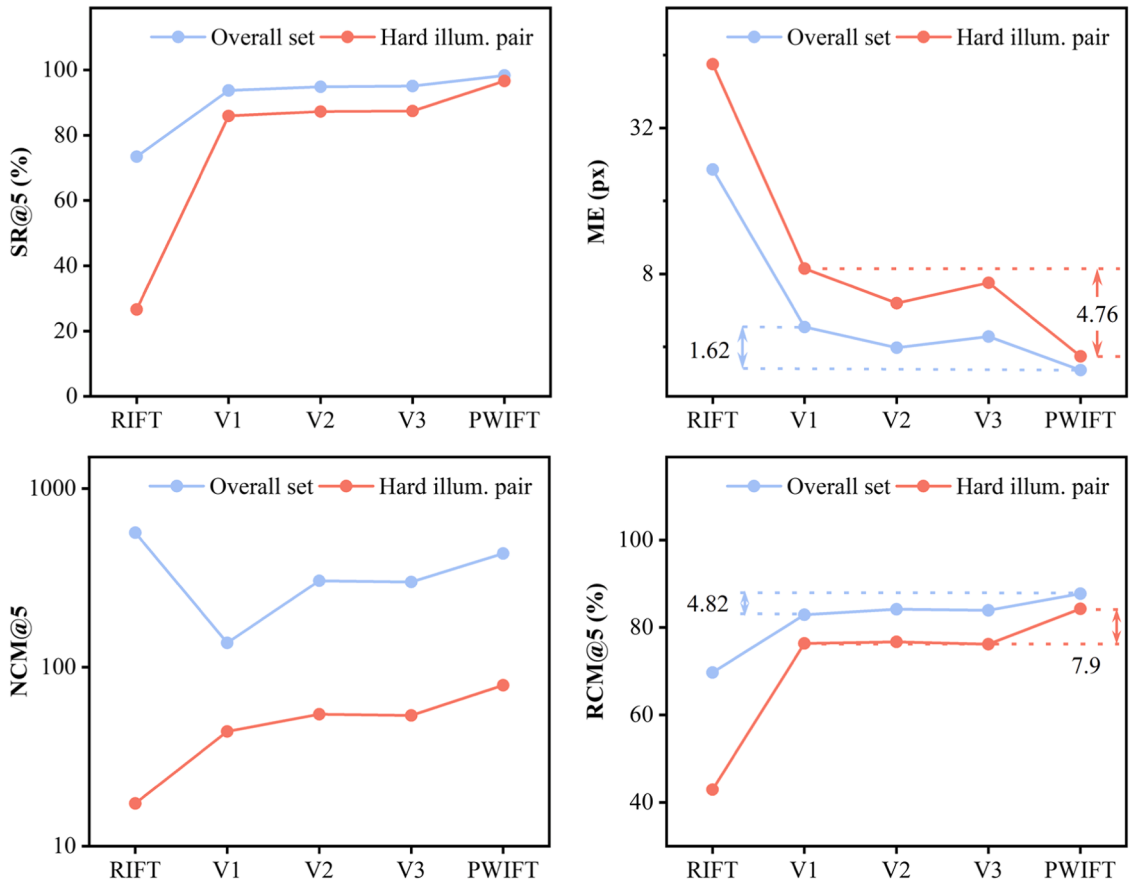

Fig. 9. Ablation trends of SR@5, ME, NCM@5, and RCM@5 for all pairs and the hard illumination pair.

introduces direct photometric reliability weighting into a RIFT-like pipeline, where the photometric weight is used for keypoint scoring and descriptor-level weighting, while the weight-guided keypoint selection strategy and the bright-dark bi-channel descriptor are not used. V2 removes the weight-guided keypoint selection from PWIFT while retaining the bright-dark bi-channel descriptor. V3 retains the weight-guided keypoint selection but replaces the bright-dark bi-channel descriptor with a single-channel weighted descriptor. The full PWIFT contains all proposed components. In this experiment, SR@5 denotes the proportion of image pairs whose mean matching error after FSC filtering is lower than 5 pixels.

The overall ablation results are shown in Table 9 and Fig. 9. Compared with RIFT, V1 increases SR@5 from 73.48% to 93.75% and reduces ME from 21.59 pixels to 4.83 pixels. This result shows that incorporating photometric reliability provides an effective prior for lunar image matching under illumination variations. Meanwhile, V1 obtains 136.79 NCM@5, whereas the full PWIFT achieves 432.29 NCM@5 and a higher RCM@5 of 87.70%. This indicates that the proposed full framework not only improves the success rate and matching error, but also produces denser reliable correspondences. V2 and V3 further demonstrate the roles of the two core components. V2 achieves 94.86% SR@5 and increases NCM@5 to 304.53, showing that the bright-dark bi-channel descriptor enhances the discriminability of feature representation and increases the number of correct matches. V3 achieves 95.09% SR@5 and 299.48 NCM@5, indicating that the weight-guided keypoint selection also contributes to robust feature localization under illumination differences. Compared with V2 and V3, the full PWIFT obtains the best overall performance, with 98.32% SR@5, 3.21 pixels ME, 432.29 NCM@5, and 87.70% RCM@5. These results suggest that the photometric weighting, weight-guided keypoint selection, and bright-dark bi-channel descriptor provide complementary improvements in matching accuracy, reliability, and correspondence density.

Table 10
Ablation results on the most challenging illumination pair M109080308R–M129133239R from E010-0.54 m Illumination.

Method	SR@5	ME	NCM@5	RCM@5
RIFT	26.63%	58.64	17.33	42.95%
V1	85.93%	8.42	43.80	76.35%
V2	87.27%	6.06	54.67	76.68%
V3	87.44%	7.36	53.80	76.15%
PWIFT	**96.65%**	**3.66**	**79.33**	**84.25%**

Table 9
Overall ablation results on the E010-0.54 m Illumination subset.

Method	SR@5	ME	NCM@5	RCM@5
RIFT	73.48%	21.59	566.59	69.70%
V1	93.75%	4.83	136.79	82.88%
V2	94.86%	3.97	304.53	84.15%
V3	95.09%	4.41	299.48	83.90%
PWIFT	**98.32%**	**3.21**	**432.29**	**87.70%**

To further examine the behavior under large illumination differences, the most challenging illumination pair, M109080308R–M129133239R, was separately evaluated, as shown in Table 10 and Fig. 9. In this case, RIFT achieves only 26.63% SR@5 and has an ME of 58.64 pixels, showing that strong illumination variation substantially affects the baseline matching performance. V1 increases SR@5 to 85.93% and reduces ME to 8.42 pixels, further confirming the effectiveness of photometric reliability weighting in difficult

Table 11
Matching results on the Ryugu-ONC dataset.

Dataset	SR@5	NCM@5	RCM@5	$RMSE_{match}$
Ryugu-ONC	100%	273.3	21.6%	3.14

illumination conditions. V2 and V3 improve SR@5 to 87.27% and 87.44%, respectively, and increase NCM@5 to >53 matches. The full PWIFT achieves the best performance on this pair, with 96.65% SR@5, 3.66 pixels ME, 79.33 NCM@5, and 84.25% RCM@5. Compared with V1, the full PWIFT further improves SR@5 by 10.72 percentage points, reduces ME by 4.76 pixels, and increases NCM@5 by 35.53. These results show that the complete PWIFT design is particularly effective for difficult illumination differences, providing more accurate and denser correspondences than the ablated variants.

4.6. Small-Body image registration test

To evaluate the applicability of PWIFT to real deep-space small-body observations, a Ryugu-ONC dataset was constructed using ONC-T sequences acquired by the Hayabusa2 mission. The dataset contains 38 image pairs selected from observations between hyb2_onc_20180630_XXX_tvf_l2d and hyb2_onc_20191107_XXX_tvf_l2d. All images were standardized to 1024 × 1024 pixels, with the small-body region located near the image center and occupying approximately one-quarter of the image. Compared with the pixel-aligned lunar benchmark, the Ryugu-ONC pairs contain a smaller target-area ratio, weaker background texture, more frequent shadow-boundary changes, brightness inversion, and moderate rotation and scale variations. These characteristics provide a complementary test for evaluating PWIFT under real panoramic small-body image registration conditions.

Table 11 reports registration results on the Ryugu-ONC dataset. PWIFT successfully registers all 38 image pairs, achieving an SR@5 of 100%, an average NCM@5 of 273.3, an RCM@5 of 21.6%, and an $RMSE_{match}$ of 3.14 pixels. Fig. 10 shows two representative examples. In Fig. 10(a), PWIFT obtains 305 correspondences with an $RMSE_{match}$ of 3.22 pixels; in Fig. 10(b), it obtains 466 correspondences with an $RMSE_{match}$ of 2.96 pixels. The green correspondence lines show that stable matches can be recovered under significant illumination differences and moderate geometric variations. The checkerboard mosaics and zoomed-in regions show continuous transitions along the small-body boundary and surface textures, without obvious tearing or local misalignment.

These results indicate that PWIFT can maintain stable inlier quality and global alignment continuity on real small-body observations. The performance is mainly supported by the illumination-field-integrated phase-congruency representation, the weight-guided keypoint detection strategy, and the bright-dark bi-channel descriptor, which jointly improve structural stability, sampling efficiency in matchable regions, and descriptor consistency under shadow migration and brightness inversion. Therefore, PWIFT is applicable not only to pixel-aligned lunar multi-illumination images, but also to real panoramic small-body image registration scenarios with more complex illumination and geometric variations.

5. Conclusion

This paper presents PWIFT, a photometric-aware invariant feature registration method for PRSIs under complex illumination and geometric variations. PWIFT addresses feature instability and unreliable tie-point establishment caused by brightness inversion, shadow migration, spatially nonuniform irradiance, and nonlinear radiometric differences. The main idea is to convert illumination geometry into a pixel-level photometric reliability prior and propagate it through multi-scale structural feature representation, keypoint selection, bright-dark bi-channel descriptor construction, and geometry-guided match refinement. Experimental results on the lunar multi-illumination benchmark show the effectiveness of the proposed PWIFT. It achieves the highest

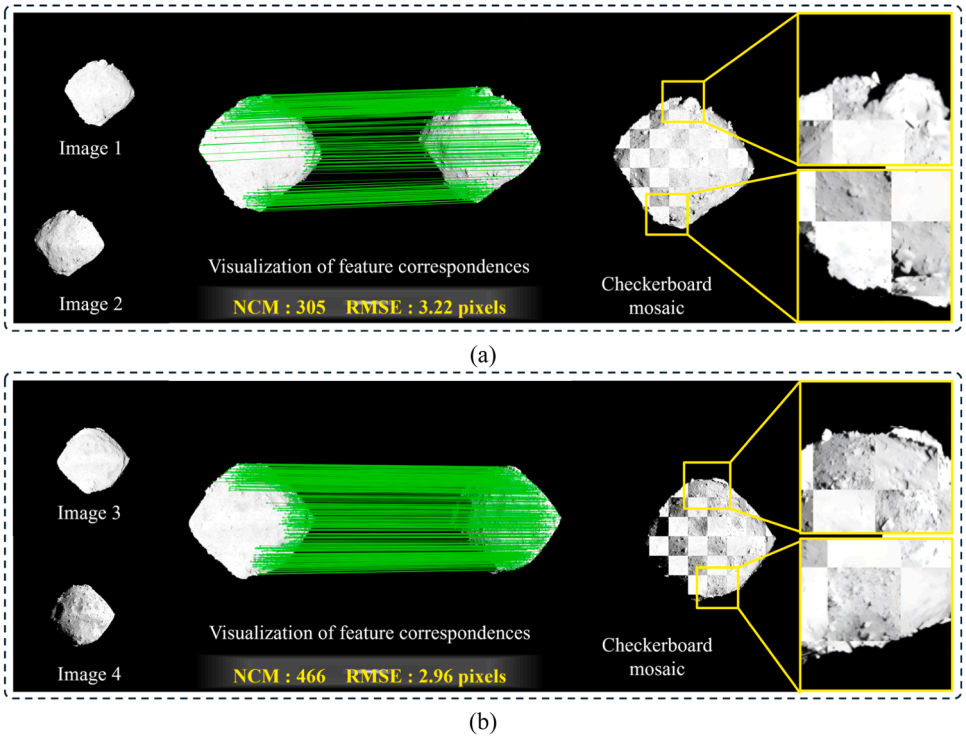

Fig. 10. Registration examples on the Ryugu-ONC small-body dataset. (a) and (b) show two representative image pairs. From left to right, we present the input images, the visualization of feature correspondences, and the checkerboard mosaics after registration with zoomed-in details for visual assessment of alignment continuity and edge consistency.

registration success rate among other evaluated methods, with an SR@5 of 97.93%, an ME@10 of 2.66 pixels, and an RCM@5 of 90.80%. The results indicate that PWIFT provides a favorable balance between registration stability, geometric accuracy, and correspondence reliability, rather than simply maximizing the number of matches.

Further analyses under different illumination, terrain, density–accuracy, ablation, and geometric perturbation settings confirm the effectiveness and robustness of the proposed design. The photometric model comparison supports the use of the Akimov disk function as the default weighting model, while the ablation results confirm the complementary contributions of photometric reliability weighting, weight-guided keypoint selection, and the bright-dark bi-channel descriptor. The successful registration of real Ryugu-ONC small-body images further provides preliminary evidence for the applicability of PWIFT beyond the lunar benchmark. Nevertheless, PWIFT still has two main limitations. First, it relies on photometric reliability maps derived from auxiliary illumination and observation geometry, and inaccurate or unavailable geometry may reduce the reliability of the weight map. Second, its current geometric refinement is mainly based on a homography model, which may be insufficient under strong relief-induced parallax, highly non-planar terrain, extremely small overlaps, or severe viewpoint changes. Future work will focus on adaptive photometric reliability estimation under incomplete auxiliary geometry and more flexible geometric models for complex planetary terrain.

CRediT authorship contribution statement

Qingyuan Yan: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis. **Yufei Guo:** Visualization, Validation, Data curation. **Xiangyuan Zeng:** Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] H. Chen, X. Hu, K. Willner, Z. Ye, F. Damme, P. Gläser, Y. Zheng, X. Tong, H. Hußmann, J. Oberst, Neural implicit shape modeling for small planetary bodies from multi-view images using a mask-based classification sampling strategy, ISPRS J. Photogramm. Remote Sens. 212 (2024) 122–145, https://doi.org/10.1016/j.isprsjprs.2024.04.029.

[2] A. Neesemann, S. van Gasselt, R. Jaumann, J.C. Castillo-Rogez, C.A. Raymond, S.H. G. Walter, F. Postberg, Improved stereophotogrammetric and multi-view shape-from-shading DTMs of Occator crater and its interior cryovolcanism-related bright spots, Remote Sens. 17 (2025) 437, https://doi.org/10.3390/rs17030437.

[3] V. Muralidharan, M.R. Makhdoomi, A. Žinys, B. Razgus, M. Klimavičius, M. Olivares-Mendez, C. Martinez, On-ground validation of orbital GNC: visual navigation assessment in robotic testbed facility, Astrodyn 9 (2025) 343–367, https://doi.org/10.1007/s42064-024-0198-4.

[4] N. Rothenberger, G. Georgakis, Y. Cheng, A. Ansar, Illumination invariant image matching for lunar TRN. AIAA SCITECH 2025 Forum, American Institute of Aeronautics and Astronautics, Orlando, FL, USA, 2025, https://doi.org/10.2514/6.2025-2073.

[5] M. Dor, T. Driver, K. Getzandanner, P. Tsiotras, AstroSLAM: autonomous monocular navigation in the vicinity of a celestial small body—theory and experiments, Int. J. Robot. Res. 43 (2024) 1770–1808, https://doi.org/10.1177/02783649241234367.

[6] X. Lin, T. Li, B. Hua, L. Li, C. Zhao, A review of deep space image-based navigation methods, Aerospace 12 (2025) 789, https://doi.org/10.3390/aerospace12090789.

[7] A.T. Fragoso, C.T. Lee, A.S. McCoy, S.-J. Chung, A seasonally invariant deep transform for visual terrain-relative navigation, Sci. Robot. 6 (2021) eabf3320, https://doi.org/10.1126/scirobotics.abf3320.

[8] S. Silvestrini, M. Piccinin, G. Zanotti, A. Brandonisio, I. Bloise, L. Feruglio, P. Lunghi, M. Lavagna, M. Varile, Optical navigation for lunar landing based on convolutional neural network crater detector, Aerosp. Sci. Technol. 123 (2022) 107503, https://doi.org/10.1016/j.ast.2022.107503.

[9] Y. Zhang, G. Wan, D. Liu, T. Tao, C. Xiao, Z. Ye, X. Tong, Preliminary validation of multimodal feature matching method for multi-source DEM registration in planetary scenarios, Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci. (2025) 1727–1732, https://doi.org/10.5194/isprs-archives-XLVIII-G-2025-1727-2025.

[10] L. Xue, Z. Ye, D. Liu, S. Liu, R. Huang, H. Xie, Y. Feng, B. Guo, Y. Xu, X. Tong, Effective feature matching of high-resolution planetary orbiter images based on optimized image partitioning and rapid local correspondence, Planet. Space Sci. 260 (2025) 106091, https://doi.org/10.1016/j.pss.2025.106091.

[11] G. Wan, R. Huang, Y. Xu, Z. Ye, Q. You, X. Yan, X. Tong, Efficient phase congruency-based feature transform for rapid matching of planetary remote sensing images, IEEE Geosci. Remote Sens. Lett. 22 (2025) 1–5, https://doi.org/10.1109/LGRS.2024.3510794.

[12] Z. Geng, B. Yang, Y. Pi, Y. Yang, Y. Zhao, FRIM: a framework for satellite-borne remote sensing image matching, IEEE Trans. Geosci. Remote Sens. 63 (2025) 1–18, https://doi.org/10.1109/TGRS.2025.3599378.

[13] J. Zhong, J. Yan, M. Li, J.-P. Barriot, A deep learning-based local feature extraction method for improved image matching and surface reconstruction from Yutu-2 PCAM images on the Moon, ISPRS J. Photogramm. Remote Sens. 206 (2023) 16–29, https://doi.org/10.1016/j.isprsjprs.2023.10.021.

[14] C. Xu, X. Huang, M. Li, D. Wang, Landmark database selection for vision-aided inertial navigation in planetary landing missions, Aerosp. Sci. Technol. 118 (2021) 107040, https://doi.org/10.1016/j.ast.2021.107040.

[15] H. Ge, Y. Geng, X. Ba, Y. Wang, J. Lv, Automated registration of full moon remote sensing images based on triangulated network constraints, Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci. (2024) 89–98, https://doi.org/10.5194/isprs-archives-XLVIII-2-2024-89-2024.

[16] J. Zhou, K. Wang, J. Wang, Terrain matching with effective 3D Zernike moments: spectrum analysis and algorithm, Aerosp. Sci. Technol. 170 (2026) 111504, https://doi.org/10.1016/j.ast.2025.111504.

[17] W. Huang, Q. Sun, Q. Xu, J. Ma, Z. Lv, T. Gao, A. Yu, A transfer learning-based contourlet residual-driven terrain-aware method for Arctic seafloor DEM super-resolution, ISPRS J. Photogramm. Remote Sens. 239 (2026) 736–754, https://doi.org/10.1016/j.isprsjprs.2026.06.023.

[18] P. Kovesi, Phase congruency: a low-level image invariant, Psychol. Res. 64 (2000) 136–148, https://doi.org/10.1007/s004260000024.

[19] Z. Geng, H. Liu, P. Duan, X. Wei, S. Li, Feature-based multimodal remote sensing image matching: benchmark and state-of-the-art, ISPRS J. Photogramm. Remote Sens. 229 (2025) 285–302, https://doi.org/10.1016/j.isprsjprs.2025.08.028.

[20] G. Wan, R. Huang, Y. Xu, Z. Ye, Y. Feng, H. Xie, X. Tong, An optimized bandpass filtering-based matching method for planetary remote sensing images with local topological prior, IEEE Trans. Geosci. Remote Sens. 63 (2025) 1–16, https://doi.org/10.1109/TGRS.2025.3544241.

[21] S. Zhang, B. Luo, J. Liu, Z. Fu, X. Su, S. Zhu, GLIFT: a global-to-local invariant feature transformation method for multimodal remote sensing image matching, IEEE Trans. Geosci. Remote Sens. 63 (2025) 1–19, https://doi.org/10.1109/TGRS.2025.3599445.

[22] Z. Fan, Y. Pi, M. Wang, Effective feature matching for multimodal remote sensing images via sparse sampling description, ISPRS J. Photogramm. Remote Sens. 230 (2025) 943–961, https://doi.org/10.1016/j.isprsjprs.2025.10.012.

[23] B. Zhu, W. Chen, J. Cao, M. Wang, X. Liu, Y. Guo, B.-H. Tang, MISR: robust multimodal remote image matching of Earth and planetary using modality-invariant structural representation, IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 19 (2026) 17877–17893, https://doi.org/10.1109/JSTARS.2026.3694934.

[24] B. Xie, B. Liu, K. Di, W.-C. Liu, Y. Kou, Y. Jia, Y. Zhang, Robust feature matching of multi-illumination lunar orbiter images based on crater neighborhood structure, Remote Sens. 17 (2025) 2302, https://doi.org/10.3390/rs17132302.

[25] R. Huang, G. Wan, Y. Zhou, Z. Ye, H. Xie, Y. Xu, X. Tong, Fast double-channel aggregated feature transform for matching planetary remote sensing images, IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 17 (2024) 9282–9293, https://doi.org/10.1109/JSTARS.2024.3390432.

[26] D. Liu, Z. Ye, Y. Xu, R. Huang, L. Xue, H. Chen, G. Wan, H. Xie, X. Tong, A mismatch removal method based on global constraint and local geometry preservation for lunar orbiter images, IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 17 (2024) 10221–10236, https://doi.org/10.1109/JSTARS.2024.3400394.

[27] D. DeTone, T. Malisiewicz, A. Rabinovich, SuperPoint: self-supervised interest point detection and description, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), IEEE, Salt Lake City, UT, USA, 2018, pp. 337–349, https://doi.org/10.1109/CVPRW.2018.00060.

[28] P. Lindenberger, P.-E. Sarlin, M. Pollefeys, LightGlue: local feature matching at light speed, in: 2023 IEEE/CVF International Conference on Computer Vision (ICCV), IEEE, Paris, France, 2023, pp. 17581–17592, https://doi.org/10.1109/ICCV51070.2023.01616.

[29] J. Sun, Z. Shen, Y. Wang, H. Bao, X. Zhou, LoFTR: detector-free local feature matching with transformers, in: 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE, Nashville, TN, USA, 2021, pp. 8918–8927, https://doi.org/10.1109/CVPR46437.2021.00881.

[30] Ö. Tuzcuoğlu, A. Köksal, B. Sofu, S. Kalkan, A.A. Alatan, XoFTR: cross-modal feature matching transformer, in: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), IEEE, Seattle, WA, USA, 2024, pp. 4275–4286, https://doi.org/10.1109/CVPRW63382.2024.00431.
[31] J. Edstedt, I. Athanasiadis, M. Wadenbäck, M. Felsberg, DKM: dense kernelized feature matching for geometry estimation, in: 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE, Vancouver, BC, Canada, 2023, pp. 17765–17775, https://doi.org/10.1109/CVPR52729.2023.01704.
[32] J. Edstedt, Q. Sun, G. Bökman, M. Wadenbäck, M. Felsberg, RoMa: robust dense feature matching, in: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE, Seattle, WA, USA, 2024, pp. 19790–19800, https://doi.org/10.1109/CVPR52733.2024.01871.
[33] J. Edstedt, D. Nordström, Y. Zhang, G. Bökman, J. Astermark, V. Larsson, A. Heyden, F. Kahl, M. Wadenbäck, M. Felsberg, RoMa v2: harder better faster denser feature matching, arXiv preprint arXiv:2511.15706 (2025). https://doi.org/10.48550/arXiv.2511.15706.
[34] T. Gao, C. Lan, W. Huang, S. Wang, SFA-net: a SAM-guided focused attention network for multimodal remote sensing image matching, ISPRS J. Photogramm. Remote Sens. 223 (2025) 188–206, https://doi.org/10.1016/j.isprsjprs.2025.02.032.
[35] T. Gao, C. Lan, C. Zhou, Y. Zhang, W. Huang, Y. Wang, L. Wang, Arctic sea ice motion retrieval from multisource SAR images using a keypoint-free feature tracking algorithm, ISPRS J. Photogramm. Remote Sens. 230 (2025) 258–274, https://doi.org/10.1016/j.isprsjprs.2025.09.013.
[36] S. Fu, Z. Xu, D. Wu, S. Gong, Deep neural network-based high-precision identification of weak stability boundary structures, IEEE Trans. Aerosp. Electron. Syst. 62 (2026) 7208–7223, https://doi.org/10.1109/TAES.2026.3668754.
[37] Z. Li, B. Wu, Semantic-aware image matching for large-scale 3-D reconstruction of the martian surface from rover images, IEEE Trans. Geosci. Remote Sens. 63 (2025) 1–17, https://doi.org/10.1109/TGRS.2025.3594413.
[38] S. Zhang, Y. Xue, Y. Tang, R. Zhu, X. Jiang, C. Niu, W. Yin, Feature scalar field grid-guided optical-flow image matching for multi-view images of asteroid, Remote Sens. 15 (2023) 5786, https://doi.org/10.3390/rs15245786.
[39] Y. Wu, W. Ma, M. Gong, L. Su, L. Jiao, A novel point-matching algorithm based on fast sample consensus for image registration, IEEE Geosci. Remote Sens. Lett. 12 (2015) 43–47, https://doi.org/10.1109/LGRS.2014.2325970.
[40] J. Li, Q. Hu, M. Ai, RIFT: multi-modal image matching based on radiation-variation insensitive feature transform, IEEE Trans. Image Process. 29 (2020) 3296–3310, https://doi.org/10.1109/TIP.2019.2959244.
[41] G. Wan, Z. Ye, Y. Xu, R. Huang, Y. Zhou, H. Xie, X. Tong, Multimodal remote sensing image matching based on weighted structure saliency feature, IEEE Trans. Geosci. Remote Sens. 62 (2024) 1–16, https://doi.org/10.1109/TGRS.2023.3347259.
[42] D.L. Domingue, B.W. Denevi, S.L. Murchie, C.D. Hash, Application of multiple photometric models to disk-resolved measurements of Mercury's surface: insights into Mercury's regolith characteristics, Icarus 268 (2016) 172–203, https://doi.org/10.1016/j.icarus.2015.11.040.
[43] K. Muinonen, K. Lumme, Disk-integrated brightness of a Lommel-Seeliger scattering ellipsoidal asteroid, Astron. Astrophys. 584 (2015) A23, https://doi.org/10.1051/0004-6361/201526456.
[44] M. Minnaert, The reciprocity principle in lunar photometry, Astrophys. J. 93 (1941) 403, https://doi.org/10.1086/144279.
[45] S.E. Schröder, J.-Y. Li, M.D. Rayman, S.P. Joy, C.A. Polanskey, U. Carsenty, J. C. Castillo-Rogez, M. Ciarniello, R. Jaumann, A. Longobardo, L.A. McFadden, S. Mottola, M. Sykes, C.A. Raymond, C.T. Russell, Ceres' opposition effect observed by the Dawn framing camera, Astron. Astrophys. 620 (2018) A201, https://doi.org/10.1051/0004-6361/201833596.